MTH321

Numerical analysis of PDEs

Mathematical formulation and methods for mixed Poisson problems

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Abstract

We study how the right-edge closure affects finite difference solutions of a Poisson problem with Dirichlet data on three sides and Neumann or Robin data on the fourth. A backward derivative row is compared with a centred ghost-point condition eliminated into a boundary PDE row. A discrete comparison principle and a boundary-adapted barrier give mesh-independent maximum-norm estimates. Although both raw boundary defects are $O(h)$, their stability weights differ, yielding $O(h)$ and $O(h^2)$ global error bounds. A separate comparison proves a matching first-order lower bound for the exponential benchmark. Positive diagonal weights produce symmetric positive definite systems and discrete energy estimates. Six-mesh experiments confirm approximately first- and second-order convergence in the trapezoidal discrete L^2, nodal maximum and closure-dependent energy norms. Independent boundary-row, manufactured-solution and uneliminated-system checks test the assembly; tolerance sweeps and weighted CG cross-check the linear solves. At $h = 1/512$ the maximum error of the ghost closure is about 2 010 (Neumann) and 1 990 (Robin) times smaller than that of the backward closure, while its matrix has only 0.08 % more nonzeros; the ghost closure reaches every fixed accuracy target on much smaller grids, whereas the backward closure does not reach targets below 2×10^{-3} within $N \leq 512$. Independent FEM and pure-Neumann depth checks extend the verification. The conclusions distinguish proved bounds, observed rates and implementation-dependent timing.

Keywords: Poisson equation; mixed boundary conditions; finite differences; ghost points; comparison principle; error estimates; reproducibility.

1 Introduction

Steady heat conduction and electrostatic models combine prescribed values, prescribed outward derivatives and exchange conditions on different boundary segments. A numerical method must translate all of these conditions into equations consistent with its interior operator. The present report studies this issue through Direction 2 of the supplied Problem Pack: a reproducible Poisson problem on the unit square with Dirichlet values on three sides and Neumann or Robin data on the open right edge.

The direction prescribes Cartesian finite differences and two boundary closures. The first replaces the boundary PDE row by a backward derivative equation. The second eliminates a conceptual ghost point from a centred derivative equation and retains a PDE row at the boundary. Our question is how this change of boundary formulation affects the solution error, the matrix structure and the cost of achieving a named accuracy target. The two formulations differ in the equations they enforce and in their row scaling, as well as in derivative accuracy.

LeVeque [1] develops the finite difference framework for boundary value problems. Within that framework, consistency must be combined with stability before it implies convergence. For the present operators, both raw boundary rows have an $O(h)$ defect. We derive a maximum-norm stability estimate that weights the ghost boundary defect by an additional factor h, explaining its $O(h^2)$ global upper bound. The analysis uses the actual mixed boundary and its corner convention.

The exact function $u^* = e^{x+y}$ determines the forcing and boundary data independently of the assembled matrix. This permits exact error evaluation and a manufactured-solution verification study in the sense of Salari and Knupp [2]. The benchmark verifies numerical methods; it does not calibrate a model against an experimental device.

The analytical contributions are an explicit comparison principle, inverse positivity, uniform maximum-norm estimates, a sharp first-order maximum-error result for the exponential FO benchmark, and a weighted energy identity with discrete L^2 and energy bounds. The weighting is a standard diagonal symmetrisation; no new general CG algorithm is claimed. The computational study tests these results in all four boundary/closure configurations, separates algebraic and

discretisation errors on a representative grid, and reports measured cost at fixed accuracy thresholds.

Section 2 states the model and regularity assumptions. Section 3 derives the discrete systems and the algorithm that assembles them. Section 4 establishes stability and convergence, and Section 5 the weighted matrix structure and the resulting solver choice. Section 6 collects the evidence that the implementation and the linear solves can be trusted. Section 7 presents the numerical experiments and the cost comparison, and Section 9 relates the measured results to the proved statements and states their limitations.

2 Mathematical model

2.1 Domain, boundary data and regularity

We now specify the selected mixed-boundary benchmark. All variables are nondimensional. Set

$$\begin{aligned} \Omega &= (0,1)^2, \\ \Gamma_D &= (\{0\}\times[0,1]) \cup ([0,1]\times\{0,1\}), \quad \Gamma_B = \{1\}\times(0,1). \end{aligned} \tag{2.1}$$

The governing equations are

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = g & \text{on } \Gamma_D, \\ \partial_n u + \beta u = s_\beta & \text{on } \Gamma_B. \end{cases} \tag{2.2}$$

The outward normal on Γ_B is $(1,0)$, so $\partial_n u = u_x$ there. The required variants are $(\beta, s_\beta) = (0, q)$ and $(2, r)$. Every corner belongs to the Dirichlet node set. In the weak formulation, traces are understood almost everywhere, and isolated corner points have zero boundary measure.

Physical interpretation of the boundary data. For a homogeneous conductor with constant thermal conductivity $k > 0$, write the dimensional position and temperature as $\boldsymbol{X} = L(x, y)$ and $T(\boldsymbol{X}) = T_{\text{ref}} + \Delta T\, u(x, y)$, where $L > 0$ and $\Delta T > 0$. Fourier's law gives the outward heat flux

$$j_n = -k\, \partial_{n_{\boldsymbol{X}}} T = -\frac{k\Delta T}{L}\, \partial_n u.$$

Thus the Neumann datum q prescribes a scaled normal derivative: $j_n = -(k\Delta T/L)q$; $q = 0$ describes an insulated edge. For convection to a surrounding fluid at temperature T_∞, with heat-transfer coefficient $H > 0$, the boundary balance is $-k\, \partial_{n_{\boldsymbol{X}}} T = H(T - T_\infty)$. Setting $u_\infty = (T_\infty - T_{\text{ref}})/\Delta T$ yields

$$\partial_n u + \mathrm{Bi}\, u = \mathrm{Bi}\, u_\infty, \qquad \mathrm{Bi} = \frac{HL}{k}. \tag{2.3}$$

This is the standard Biot-number interpretation of a convective Robin condition [12]. Under this particular scaling, $\beta = 2$ corresponds to $\mathrm{Bi} = 2$, and the Robin load is $r = 2u_\infty$; r itself is not a heat flux. The manufactured data specify a numerical benchmark, without identifying a particular material or device.

Figure 1 fixes the boundary ownership used below.

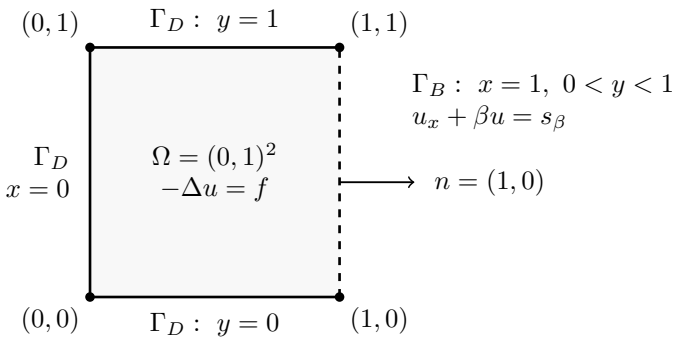


Figure 1: Boundary ownership for the selected benchmark. The solid edges and all four filled corner points carry Dirichlet data. The dashed open right edge carries the derivative/Robin condition; its endpoints are excluded from the boundary-row unknowns. This analytical schematic fixes the outward-normal sign and the corner convention before any computation.

Assumption 2.1 (Smooth solution and consistent data)**.** The coefficient $\beta \geq 0$ is constant and independent of the mesh. The exact solution u extends to a C^6 function on an open neighbourhood of $\overline{\Omega}$. The data satisfy $f = -\Delta u$ on $\overline{\Omega}$, $g = u|_{\Gamma_D}$ and $s_\beta = (u_x + \beta u)|_{\Gamma_B}$. Dirichlet values used in the analysis are imposed exactly. For $\ell = 2, 3, 4$, write

$$M_\ell = \max\{\|\partial_x^\ell u\|_{L^\infty(\overline{\Omega})}, \|\partial_y^\ell u\|_{L^\infty(\overline{\Omega})}\}. \tag{2.4}$$

All asymptotic remainders below are uniform over the relevant grid nodes; their constants may depend on derivative bounds for u, but not on N, h, i or j.

This assumption is verified for the manufactured benchmark. It is an assumption about the solution, rather than a regularity conclusion for arbitrary mixed data; general mixed-boundary problems may have corner singularities.

The manufactured data are compatible at the two right-edge endpoints, while both corner nodes remain Dirichlet nodes.

2.2 Manufactured benchmark

For the prescribed exact solution $u^*(x, y) = e^{x+y}$, direct differentiation gives

$$\begin{aligned} f(x,y) &= -2e^{x+y}, \\ g(0,y) = e^y, \qquad g(x,0) &= e^x, \qquad g(x,1) = e^{x+1}, \\ q(y) = e^{1+y}, \qquad r(y) &= 3e^{1+y}, \quad s_\beta(y) = (1+\beta)e^{1+y}. \end{aligned} \tag{2.5}$$

The corner values at $(0,0), (0,1), (1,0), (1,1)$ are $1, e, e, e^2$, respectively, and $M_2 = M_3 = M_4 = e^2$. The forcing is derived from the continuous solution independently of the discrete matrix. Consequently, substituting sampled exact values into the discrete equations retains the defects that the verification study must detect. This independently constructed benchmark permits exact error evaluation and systematic refinement [2].

2.3 Weak formulation and uniqueness

Let Tr denote the Sobolev trace and set

$$V_0 = \{v \in H^1(\Omega) : \operatorname{Tr} v = 0 \text{ on } \Gamma_D\}, \qquad V_g = \{v \in H^1(\Omega) : \operatorname{Tr} v = g \text{ on } \Gamma_D\}.$$

For $u \in V_g$, the weak formulation is

$$a_\beta(u, v) = \ell(v) \qquad (v \in V_0), \tag{2.6}$$

where

$$a_\beta(u,v) = \int_\Omega \nabla u \cdot \nabla v \, dx\, dy + \beta \int_{\Gamma_B} uv \, ds, \tag{2.7}$$
$$\ell(v) = \int_\Omega f v \, dx\, dy + \int_{\Gamma_B} s_\beta v \, ds.$$

The boundary values in these integrals denote traces. This sign convention agrees with the standard weak treatment of Neumann and Robin data in [3].

Lemma 2.2 (Coercivity induced by the left Dirichlet edge)**.** *For every $v \in V_0$,*

$$\|v\|^2_{L^2(\Omega)} \le \tfrac{1}{2}\|v_x\|^2_{L^2(\Omega)}, \qquad \|\operatorname{Tr} v\|^2_{L^2(\Gamma_B)} \le \|v_x\|^2_{L^2(\Omega)}. \tag{2.8}$$

Consequently, for $\beta \ge 0$,

$$a_\beta(v,v) \ge \tfrac{2}{3}\|v\|^2_{H^1(\Omega)}, \qquad \|v\|^2_{H^1(\Omega)} := \|v\|^2_{L^2(\Omega)} + \|\nabla v\|^2_{L^2(\Omega)}. \tag{2.9}$$

Proposition 2.3 (Solvability for the stated smooth data)**.** *Under Assumption 2.1, the function u is the unique weak solution of (2.6). In particular, both required benchmark variants have the unique solution $u^* = e^{x+y}$.*

The left Dirichlet edge gives the Poincaré inequality in Lemma 2.2, so a_β is coercive for $\beta \ge 0$. Green's identity verifies existence for the manufactured solution, while coercivity gives uniqueness. Complete proofs are given in Appendix A.

The Dirichlet boundary removes the constant nullspace even when $\beta = 0$. A pure-Neumann compatibility condition and a pressure gauge are therefore not needed for this problem. No existence claim for arbitrary nonsmooth data is used in the subsequent analysis.

3 Finite difference discretisation

3.1 Grid, unknowns and norms

For $N \ge 2$, let $h = 1/N$, $x_i = ih$ and $y_j = jh$. Define the full index set $\mathcal{X}_h = \{0,\dots,N\}^2$ and the disjoint node sets

$$\begin{aligned} \mathcal{I}_h &= \{(i,j) : 1 \le i,j \le N-1\}, \\ \mathcal{B}_h &= \{(N,j) : 1 \le j \le N-1\}, \\ \mathcal{U}_h &= \mathcal{I}_h \cup \mathcal{B}_h, \qquad\qquad \mathcal{D}_h = \mathcal{X}_h \setminus \mathcal{U}_h. \end{aligned} \tag{3.1}$$

A grid function is a real array on $\mathcal{X}_h$. A vector on $\mathcal{U}_h$ will be extended by zero on $\mathcal{D}_h$ whenever used in an error estimate or quadratic form. The ordering is

$$k(i,j) = (i-1) + N(j-1), \qquad 0 \le k < N(N-1) = n_{\mathrm{dof}}. \tag{3.2}$$

There are $(N-1)^2$ interior and $N-1$ right-edge unknowns. Ghost values are not degrees of freedom.

For a full grid function V, write

$$\begin{aligned} \|V\|_{\infty,h} &= \max_{(i,j)\in\mathcal{X}_h} |V_{i,j}|, \\ \|V\|^2_{2,h} &= h^2 \sum_{i=0}^{N}\sum_{j=0}^{N} \omega_i\omega_j V^2_{i,j}, \qquad \omega_0 = \omega_N = \tfrac{1}{2}, \quad \omega_i = 1 \ (1 \le i \le N-1). \end{aligned} \tag{3.3}$$

For arrays on the interior and open right edge, respectively, also define

$$\|F\|^2_{I,h} = h^2 \sum_{(i,j)\in\mathcal{I}_h} F^2_{i,j}, \quad \|S\|^2_{\Gamma,h} = h \sum_{j=1}^{N-1} S^2_j. \tag{3.4}$$

Subscripts ∞, I and ∞, B will denote maxima over $\mathcal{I}_h$ and $\mathcal{B}_h$.

3.2 Interior stencil and first-order closure

The centred second differences are

$$(D^h_{xx}V)_{i,j} = \frac{V_{i+1,j} - 2V_{i,j} + V_{i-1,j}}{h^2}, \qquad (D^h_{yy}V)_{i,j} = \frac{V_{i,j+1} - 2V_{i,j} + V_{i,j-1}}{h^2}.$$

The usual Taylor derivation of these formulas is recalled and quantified in Lemma 4.1; see also [4]. The interior operator is $L_h = -D^h_{xx} - D^h_{yy}$, giving

$$(L_hU)_{i,j} = \frac{4U_{i,j} - U_{i-1,j} - U_{i+1,j} - U_{i,j-1} - U_{i,j+1}}{h^2} = f_{i,j}, \qquad (i,j) \in \mathcal{I}_h. \tag{3.5}$$

Its neighbour at $i = N$ remains an unknown.

The first-order (FO) closure replaces the boundary PDE row with

$$(B^\beta_h U)_j = \left(\frac{1}{h} + \beta\right) U_{N,j} - \frac{1}{h}U_{N-1,j} = s_{\beta,j}, \qquad 1 \le j \le N-1. \tag{3.6}$$

This equation has no tangential coupling. Its diagonal is $1/h$ for Neumann data and $1/h + 2$ for the prescribed Robin data.

3.3 Ghost-point closure

Define the eliminated ghost (G) operator by

$$(G^\beta_h V)_j = \frac{4V_{N,j} - 2V_{N-1,j} - V_{N,j-1} - V_{N,j+1}}{h^2} + \frac{2\beta}{h}V_{N,j}. \tag{3.7}$$

Proposition 3.1 (Exact ghost elimination)**.** *For a fixed boundary index j, the centred boundary condition*

$$\frac{U_{N+1,j} - U_{N-1,j}}{2h} + \beta U_{N,j} = s_{\beta,j} \tag{3.8}$$

and the boundary five-point PDE row

$$4U_{N,j} - U_{N-1,j} - U_{N+1,j} - U_{N,j-1} - U_{N,j+1} = h^2 f_{N,j} \tag{3.9}$$

are equivalent to the single retained equation

$$(G^\beta_h U)_j = f_{N,j} + \frac{2s_{\beta,j}}{h} \tag{3.10}$$

together with the uniquely determined auxiliary value

$$U_{N+1,j} = U_{N-1,j} + 2hs_{\beta,j} - 2h\beta U_{N,j}. \tag{3.11}$$

Proof. Since $h > 0$, multiplying (3.8) by $2h$ and solving for its ghost value gives (3.11). Substitution into (3.9) gives

$$(4 + 2h\beta)U_{N,j} - 2U_{N-1,j} - U_{N,j-1} - U_{N,j+1} = h^2 f_{N,j} + 2hs_{\beta,j}.$$

Dividing by h^2 proves (3.10). Conversely, given retained values satisfying (3.10), define the ghost value by (3.11). The centred boundary condition then holds identically, and multiplication of (3.10) by h^2 and reverse substitution recovers (3.9). This proves both directions and uniqueness of the auxiliary value. □

Thus the ghost method imposes a boundary PDE row after elimination, whereas the first method imposes a derivative row. Both have exactly $N(N-1)$ retained equations and unknowns.

3.4 Dirichlet elimination and matrix scaling

For $c \in \{\mathrm{FO}, \mathrm{G}\}$, let A_h^c denote the reduced matrix obtained from (3.5) and the corresponding raw boundary row (3.6) or (3.10). If $\mathcal{A}_h^c$ is the stencil matrix with columns for all nodes, the reduced system is

$$A_h^c \boldsymbol{U}_h^c = \boldsymbol{b}_h^c, \qquad A_h^c = (\mathcal{A}_h^c)_{\mathcal{U}_h \mathcal{U}_h}, \qquad \boldsymbol{b}_h^c = \boldsymbol{F}_h^c - (\mathcal{A}_h^c)_{\mathcal{U}_h \mathcal{D}_h} \boldsymbol{g}_h. \tag{3.12}$$

Here $\boldsymbol{F}_h^c$ is f in the interior and is s_β (FO) or $f + 2s_\beta/h$ (G) on the right edge. In particular, a known entry $a_{kD} g_D$ contributes $-a_{kD} g_D$ to the right-hand side. The ghost rows at $j = 1$ and $j = N - 1$ receive $g(1, 0)/h^2$ and $g(1, 1)/h^2$, respectively. For $N = 2$ both contributions enter the same row. Neither corner is assigned a derivative row.

Across the interface between $\mathcal{I}_h$ and $\mathcal{B}_h$, the interior coefficient is $-1/h^2$, while the reverse coefficient is $-1/h$ (FO) or $-2/h^2$ (G). Hence both raw matrices are nonsymmetric for $N \geq 2$. Their symmetrised forms will be established in Proposition 5.1, rather than inferred from the continuous Laplacian.

Proposition 3.2 (Analytical sparsity counts)**.** *For $N \geq 2$ and $\beta \geq 0$, the numbers of nonzero entries in the reduced matrices are*

$$\begin{aligned} \operatorname{nnz}(A_h^{\mathrm{FO}}) &= (N-1)(5N-6), \\ \operatorname{nnz}(A_h^{\mathrm{G}}) &= (N-1)(5N-6) + 2(N-2). \end{aligned} \tag{3.13}$$

Both counts grow linearly with $n_{\mathrm{dof}} = N(N-1)$.

Proof sketch. Count the entries row by row: every interior row has a diagonal and its east, west and vertical unknown neighbours; every boundary row has a diagonal and a west entry; the ghost closure adds the $2(N-2)$ tangential boundary couplings. The complete proof is given in Appendix A. □

For both $\beta = 0$ and $\beta = 2$, the assembled matrices in the four-grid refinement data have exactly the nonzero counts in (3.13): the FO/G pairs are $(1110, 1138)$ at $N = 16$, $(4774, 4834)$ at $N = 32$, $(19782, 19906)$ at $N = 64$, and $(80518, 80770)$ at $N = 128$. This agreement checks that the implemented connectivity matches the analytical stencil counts. A nonzero-count check alone does not verify the coefficient values or the right-hand side. The counts refer to the reduced matrices before factorisation.

3.5 Algorithm and implementation

The project-specific code implements grid ownership, the ordering (3.2), both boundary closures, exact Dirichlet elimination, error norms, componentwise residuals, row weights and the edge-sum energy. Sparse entries are assembled using array masks and vectorised coordinate construction in COO and converted to CSR; the direct solve converts the matrix to CSC. Assembly, reconstruction, exact-vector evaluation and row-weight construction contain no Python loops over grid points. The discretisation, boundary closures, Dirichlet elimination, error diagnostics and weighted structure are implemented by the authors, because they are the object of the verification. NumPy supplies array operations and a dense small-system solve; SciPy [20] supplies sparse storage, SuperLU, GMRES, ILU, CG and the symmetric eigenvalue routine; Matplotlib supplies figures. These interfaces are delegated numerical services rather than separately implemented solvers: the aim is to test the discretisation, not to reimplement sparse factorisation or Krylov methods.

The verification of this algorithm is the subject of Section 6; the experiments that use it are reported in Section 7.

1. **Input:** $N \geq 2$, closure $c \in \{\mathrm{FO}, \mathrm{G}\}$, $\beta \in \{0, 2\}$, and independent functions u^*, f, s_β. Set $h = 1/N$ and enumerate $\mathcal{U}_h$ by (3.2).
2. For each interior node, assemble (3.5). For each open right-edge node, assemble (3.6) or (3.10). Do not assign a derivative row to a corner.
3. For each known Dirichlet neighbour with coefficient a_{kD}, add $-a_{kD}g_D$ to the right-hand side. Store only unknown columns.
4. Convert COO entries to CSR, sum duplicate entries, and check the dimensions and finiteness of $A_h^c, \boldsymbol{b}_h^c$.
5. Solve $A_h^c \widetilde{\boldsymbol{U}}_h^c = \boldsymbol{b}_h^c$ using the fixed sparse direct backend. Reconstruct all grid values, retaining exact Dirichlet values.
6. Evaluate the global norms (3.3), restricted boundary/interior errors, residuals (6.1), matrix asymmetry and, where required, the weighted energy error.
7. **Output:** solution, diagnostics and measured times. The manufactured error is evaluated against u^*, not inferred from the residual.

Algorithm 3.1: A mixed-boundary finite difference solve and its diagnostics

4 Stability and error analysis

4.1 Uniform consistency estimates

Let u_h denote u sampled on the full grid and let $\boldsymbol{u}_h$ be its restriction to the unknowns. Define

$$\boldsymbol{\tau}_h^c = A_h^c \boldsymbol{u}_h - \boldsymbol{b}_h^c. \tag{4.1}$$

Exact Dirichlet elimination means that these entries are precisely

$$\begin{aligned} \tau_{I,i,j} &= (L_h u_h)_{i,j} - f_{i,j}, \\ \tau_{B,j}^{\mathrm{FO}} &= (B_h^\beta u_h)_j - s_{\beta,j}, \\ \tau_{B,j}^{\mathrm{G}} &= (G_h^\beta u_h)_j - f_{N,j} - 2s_{\beta,j}/h. \end{aligned} \tag{4.2}$$

This definition fixes the row scaling in every subsequent estimate.

Lemma 4.1 (Local defects and uniform bounds)**.** *Under Assumption 2.1, the following expansions hold, with derivatives evaluated at the relevant node:*

$$\tau_I = -\frac{h^2}{12}(u_{xxxx} + u_{yyyy}) + O(h^4), \tag{4.3}$$

$$\tau_B^{\mathrm{FO}} = -\frac{h}{2}u_{xx} + \frac{h^2}{6}u_{xxx} + O(h^3), \tag{4.4}$$

$$\tau_B^{\mathrm{G}} = \frac{h}{3}u_{xxx} - \frac{h^2}{12}(u_{xxxx} + u_{yyyy}) + O(h^3). \tag{4.5}$$

For every $N \geq 2$, the explicit bounds are

$$\begin{aligned} \|\tau_I\|_{\infty,I} &\leq \frac{M_4}{6}h^2, \\ \|\tau_B^{\mathrm{FO}}\|_{\infty,B} &\leq \frac{M_2}{2}h, \\ \|\tau_B^{\mathrm{G}}\|_{\infty,B} &\leq \frac{M_3}{3}h + \frac{M_4}{12}h^2. \end{aligned} \tag{4.6}$$

Proof sketch. Taylor expansion of u along the grid lines. Fourth-order expansions of the centred differences and $f = -\Delta u$ give the interior defect; the second-order Taylor remainder of $u(1-h, y_j)$ gives the FO defect; for G the inward and tangential differences are expanded separately, the third-order remainder giving $M_3 h/3$ and the centred tangential defect $M_4 h^2/12$. The complete proof is given in Appendix A. □

Corollary 4.2 (Exact benchmark identities)**.** *For $u^* = e^{x+y}$, the defects satisfy*

$$\begin{aligned} \frac{\tau_I}{u^*} &= 2 + \frac{4 - 2e^h - 2e^{-h}}{h^2}, \\ \frac{\tau_B^{\mathrm{FO}}}{u^*} &= \frac{1 - e^{-h}}{h} - 1, \\ \frac{\tau_B^{\mathrm{G}}}{u^*} &= \frac{4 - 3e^{-h} - e^h}{h^2} - \frac{2}{h} + 2. \end{aligned} \tag{4.7}$$

The limits of these ratios divided by h^2, h and h, respectively, are $-1/6$, $-1/2$ and $1/3$ as $h \to 0$.

Proof. Relative to a node's exact value, its east and north neighbours contribute e^h, while its west and south neighbours contribute e^{-h}. Insert these ratios and the data (2.5) into (4.2). The Robin terms cancel exactly. Expanding $e^{\pm h}$ through the first nonzero defect term gives the stated limits. □

Although the centred derivative in (3.8) has an $O(h^2)$ defect, the raw eliminated PDE row has an $O(h)$ defect. The global effect of that row must be established using its boundary stability weight.

4.2 Comparison principle and discrete solvability

Lemma 4.3 (Discrete comparison principle)**.** *Let $N \geq 2$ and $\beta \geq 0$. Suppose a grid function V satisfies $V \geq 0$ on $\mathcal{D}_h$ and $L_h V \geq 0$ on $\mathcal{I}_h$. If either $B_h^\beta V \geq 0$ on $\mathcal{B}_h$ or $G_h^\beta V \geq 0$ there, then $V \geq 0$ on the whole grid.*

Proof sketch. If the minimum $m < 0$ is attained at an interior node, the interior inequality forces all four neighbours to equal m; repeating along decreasing i reaches a Dirichlet node with nonnegative value, a contradiction. At a right-edge node the FO or G row gives a strict contradiction for $\beta > 0$, and for $\beta = 0$ forces the interior west neighbour to be a minimum as well. The complete proof is given in Appendix A. □

We use the standard inverse-positive characterisation of nonsingular M-matrices; see Berman and Plemmons [5]. The properties needed for these particular matrices are proved directly below.

Proposition 4.4 (Invertibility and inverse positivity)**.** *For $N \geq 2$, $\beta \geq 0$ and either closure, A_h^c is invertible, has nonpositive off-diagonal entries, and satisfies $(A_h^c)^{-1} \geq 0$ entrywise. Thus it has the inverse-positive characterisation of a nonsingular M-matrix.*

Proof sketch. The off-diagonal signs follow from the stencils. If $A_h^c V = 0$, Lemma 4.3 applied to $\pm V$ gives $V = 0$, so A_h^c is invertible. For $d \geq 0$ the solution of $A_h^c V = d$ is nonnegative by the same lemma, so every column of $(A_h^c)^{-1}$ is nonnegative. The complete proof is given in Appendix A. □

4.3 Uniform maximum-norm stability

Lemma 4.5 (A boundary-adapted barrier). *Let $\psi_{i,j} = \psi(x_i)$, where $\psi(x) = 1 + x - x^2/4$. For $N \geq 2$, $1 \leq \psi_{i,j} \leq 7/4$, and the full-grid operators satisfy*

$$\begin{aligned} L_h\psi &= \tfrac{1}{2}, \\ B_h^\beta\psi &= \tfrac{1}{2} + \tfrac{h}{4} + \tfrac{7\beta}{4} \geq \tfrac{1}{2}, \\ G_h^\beta\psi &= \tfrac{1}{h} + \tfrac{1}{2} + \tfrac{7\beta}{2h} \geq \tfrac{1}{h}. \end{aligned} \tag{4.8}$$

Proof. Since $\psi'(x) = 1 - x/2 > 0$ on $[0,1]$, its endpoint values give the stated range. The centred differences are exact on this quadratic and ψ does not depend on y, so $L_h\psi = -\psi'' = 1/2$. Moreover $\psi(1) = 7/4$ and $\psi(1) - \psi(1-h) = h/2 + h^2/4$. Substitution into the FO row yields the second identity. In the ghost row the tangential differences vanish, leaving $2[\psi(1) - \psi(1-h)]/h^2 + 2\beta\psi(1)/h$, which gives the third identity. The inequalities use $\beta \geq 0$. □

Proposition 4.6 (Uniform stability in the raw row scaling). *Let $N \geq 2$, $\beta \geq 0$ and $V = 0$ on $\mathcal{D}_h$. Suppose $L_hV = F$ on $\mathcal{I}_h$ and the selected boundary operator applied to V equals S on $\mathcal{B}_h$. Set $F_0 = \|F\|_{\infty,I}$ and $S_0 = \|S\|_{\infty,B}$. Then*

$$\|V\|_{\infty,h} \leq \tfrac{7}{2}(F_0 + S_0) \qquad \text{for FO,} \tag{4.9}$$

$$\|V\|_{\infty,h} \leq \tfrac{7}{4}(2F_0 + hS_0) \qquad \text{for G.} \tag{4.10}$$

The constants are independent of N and $\beta \geq 0$.

Proof. Use the full-grid barrier of Lemma 4.5. For FO, set $a = 2(F_0 + S_0)$. On $\mathcal{D}_h$, $a\psi \pm V = a\psi \geq 0$. In the interior, $L_h(a\psi \pm V) = a/2 \pm F \geq a/2 - F_0 \geq 0$. On the right edge, $B_h^\beta(a\psi \pm V) \geq a/2 - S_0 \geq 0$. Lemma 4.3 gives $a\psi - V \geq 0$ and $a\psi + V \geq 0$, so $|V| \leq a\psi \leq 7a/4$, proving (4.9).

For G, choose $a = 2F_0 + hS_0$. The Dirichlet values are again nonnegative, the interior rows satisfy $a/2 - F_0 = hS_0/2 \geq 0$, and the boundary rows satisfy $G_h^\beta(a\psi \pm V) \geq a/h - S_0 = 2F_0/h \geq 0$. The same comparison argument yields $|V| \leq 7a/4$ and proves (4.10). The barrier has its own nonzero Dirichlet values; the proof uses full-grid operators, not a reduced matrix multiplication that discards those values. □

Theorem 4.7 (Maximum-norm convergence). *Under Assumption 2.1, let U_h^c be the unique discrete solution, including its exact Dirichlet values, and set $e_h^c = U_h^c - u_h$. For all $N \geq 2$,*

$$\|e_h^{\mathrm{FO}}\|_{\infty,h} \leq \tfrac{7}{2}\left(\tfrac{M_4}{6}h^2 + \tfrac{M_2}{2}h\right), \tag{4.11}$$

$$\|e_h^{\mathrm{G}}\|_{\infty,h} \leq \tfrac{7}{4}\left(\tfrac{M_4+M_3}{3}h^2 + \tfrac{M_4}{12}h^3\right). \tag{4.12}$$

Consequently, these maximum-norm errors are $O(h)$ and $O(h^2)$, respectively, as $N \to \infty$.

Proof. Proposition 4.4 gives existence and uniqueness of each discrete solution. On $\mathcal{D}_h$, $e_h^c = 0$. Subtraction of (4.1) from the discrete system gives

$$A_h^c \boldsymbol{e}_h^c = -\boldsymbol{\tau}_h^c. \tag{4.13}$$

Thus its full-grid interior equation is $L_h e_h^c = -\tau_I$ and its boundary equation is $B_h^\beta e_h^{\mathrm{FO}} = -\tau_B^{\mathrm{FO}}$ or $G_h^\beta e_h^{\mathrm{G}} = -\tau_B^{\mathrm{G}}$. Apply Proposition 4.6 and then Lemma 4.1 to obtain (4.11) and (4.12). Since $h \leq 1/2$ and M_ℓ is independent of h, the first bound is at most $\frac{7}{2}(M_4/12 + M_2/2)h$, and the second at most $\frac{7}{4}[(M_4 + M_3)/3 + M_4/24]h^2$. This proves uniform convergence with the stated upper orders. □

Theorem 4.8 (Sharp first-order maximum error for the exponential benchmark)**.** *Let $u^* = e^{x+y}$ and fix $\beta \geq 0$. For the exact FO discrete solution and every $N \geq 2$,*

$$\max_{1\leq j\leq N-1} |e^{\mathrm{FO}}_{N,j}| \geq C_\beta h, \qquad C_\beta := \frac{e\sinh\pi}{2\sqrt{2}\,m_\beta} > 0, \quad m_\beta := \pi\cosh\pi + \beta\sinh\pi. \tag{4.14}$$

Together with Theorem 4.7, this implies $\|e^{\mathrm{FO}}_h\|_{\infty,h} = \Theta(h)$ and $\max_j |e^{\mathrm{FO}}_{N,j}| = \Theta(h)$ for each fixed β, in particular for both required variants.

Proof sketch. The interior defect of e^{x+y} is nonpositive and, since $y_j \geq h$, $-\tau^{\mathrm{FO}}_{B,j} \geq eh/2$. The function $\phi = \sinh(\pi x)\sin(\pi y)$ has $L_h\phi \leq 0$ and $B^\beta_h\phi \leq m_\beta$ on the right edge. With $a = e/(2m_\beta)$ the grid function $V = e^{\mathrm{FO}}_h - ah\phi$ satisfies the hypotheses of Lemma 4.3, so $e^{\mathrm{FO}}_h \geq ah\phi$; evaluating at the boundary node nearest $y = 1/2$ gives $C_\beta h$. The complete proof is given in Appendix A. □

Remark 4.9 (Rates, bounds and measured errors). The factor h multiplying the ghost boundary defect in (4.10) converts its $O(h)$ local size into an $O(h^2)$ contribution to the solution error. Theorem 4.7 gives rigorous a priori upper bounds, but does not identify an asymptotic leading-error coefficient.

For the exponential benchmark, let $E^{\mathrm{FO}}_h = \|e^{\mathrm{FO}}_h\|_{\infty,h}$ and define

$$B_{\mathrm{FO}}(h) = \frac{7e^2}{2}\left(\frac{h^2}{6} + \frac{h}{2}\right).$$

Since $M_2 = M_4 = e^2$, Theorems 4.7 and 4.8 give the two-sided estimate

$$C_\beta h \leq E^{\mathrm{FO}}_h \leq B_{\mathrm{FO}}(h), \qquad C_\beta = \frac{e\sinh\pi}{2\sqrt{2}\,(\pi\cosh\pi + \beta\sinh\pi)}.$$

Here $C_0 = 0.3047737605\ldots$ and $C_2 = 0.1864919154\ldots$. At $N = 128$, the Neumann lower bound is 2.3810450×10^{-3} and the upper bound is 1.0128533×10^{-1}, whereas the measured maximum error is 6.8528917×10^{-3}. Thus the upper bound is approximately 14.8 times the measured error. Table 1 also gives the Robin comparison.

The bound is conservative because its proof replaces spatially varying defects by uniform derivative bounds and uses a common comparison barrier for the interior and boundary forcing. It remains a valid error estimate, but should not be read as a sharp prediction of the observed error magnitude. Likewise, C_β is a guaranteed lower-bound constant, rather than an identified leading coefficient. Together the bounds prove $E^{\mathrm{FO}}_h = \Theta(h)$ for this benchmark and each fixed $\beta \geq 0$. For G, the established conclusion is an $O(h^2)$ upper bound; no matching lower bound is proved here. For affine solutions the defect bounds vanish for both closures, and for quadratic solutions they vanish for G, so an upper-order statement alone does not imply a nonzero error for every smooth solution.

Boundary	C_β	$C_\beta h$	$\widehat{E}^{\mathrm{FO}}_h$	$B_{\mathrm{FO}}(h)$	$B_{\mathrm{FO}}/\widehat{E}^{\mathrm{FO}}_h$
Neumann ($\beta = 0$)	0.304774	2.38105×10^{-3}	6.85289×10^{-3}	1.01285×10^{-1}	14.8
Robin ($\beta = 2$)	0.186492	1.45697×10^{-3}	4.15872×10^{-3}	1.01285×10^{-1}	24.4

Table 1: FO maximum errors at $h = 1/128$: theoretical lower bound, measured error and theoretical upper bound. The theoretical inequalities concern the exact discrete solution. The hatted quantity is evaluated from the floating-point computed solution, using the existing refinement data; the comparison is illustrative and does not replace an algebraic-error analysis.

5 Matrix structure and solver choice

5.1 Weighted symmetry and the energy identity

For either closure define the positive diagonal matrix W_h^c by

$$(W_h^c)_{kk} = \begin{cases} 1, & k \text{ corresponds to } \mathcal{I}_h, \\ 1/h, & k \text{ corresponds to } \mathcal{B}_h,\ c = \mathrm{FO}, \\ 1/2, & k \text{ corresponds to } \mathcal{B}_h,\ c = \mathrm{G}, \end{cases} \qquad K_h^c = W_h^c A_h^c. \tag{5.1}$$

For a vector V, extended by zero on $\mathcal{D}_h$, write

$$\begin{aligned} S_x(V) &= \sum_{j=1}^{N-1} \sum_{i=0}^{N-1} (V_{i+1,j} - V_{i,j})^2, \\ S_{y,I}(V) &= \sum_{i=1}^{N-1} \sum_{j=0}^{N-1} (V_{i,j+1} - V_{i,j})^2, \\ S_{y,B}(V) &= \sum_{j=0}^{N-1} (V_{N,j+1} - V_{N,j})^2. \end{aligned} \tag{5.2}$$

Proposition 5.1 (Symmetry, energy identity and positive definiteness)**.** *For $N \geq 2$ and $\beta \geq 0$, K_h^c is real symmetric positive definite. For every real vector $\boldsymbol{V}$,*

$$h^2 \boldsymbol{V}^T K_h^c \boldsymbol{V} = S_x(V) + S_{y,I}(V) + \theta_c S_{y,B}(V) + \beta h \sum_{j=1}^{N-1} V_{N,j}^2, \quad \theta_{\mathrm{FO}} = 0, \quad \theta_{\mathrm{G}} = \tfrac{1}{2}. \tag{5.3}$$

Proof sketch. Multiplying the boundary rows by the weights makes each interior–boundary coefficient pair equal, so K_h^c is symmetric. Summation by parts over horizontal and vertical grid lines gives the energy identity, whose right-hand side is nonnegative and vanishes only for $V = 0$. The complete proof is given in Appendix A. □

It follows that

$$\|V\|_{E,h,c} = \left(h^2 \boldsymbol{V}^T K_h^c \boldsymbol{V}\right)^{1/2} \tag{5.4}$$

is a norm on the unknown vectors with zero Dirichlet extension.

5.2 Discrete coercivity and energy-error bounds

Lemma 5.2 (Discrete Poincaré and right-edge trace bounds)**.** *For $V = 0$ on $\mathcal{D}_h$, $N \geq 2$ and $\beta \geq 0$,*

$$\|V\|_{2,h}^2 \leq S_x(V) \leq \|V\|_{E,h,c}^2, \qquad \|V_B\|_{\Gamma,h}^2 \leq S_x(V) \leq \|V\|_{E,h,c}^2. \tag{5.5}$$

Proof sketch. Write $V_{i,j}$ as a telescoping sum of horizontal differences, apply the Cauchy–Schwarz inequality with $V_{0,j} = 0$, and use $hN = 1$. The complete proof is given in Appendix A. □

Proposition 5.3 (Uniform energy stability)**.** *Under the hypotheses of Proposition 4.6,*

$$\|V\|_{E,h,c} \leq \|F\|_{I,h} + \gamma_c \|S\|_{\Gamma,h}, \qquad \gamma_{\mathrm{FO}} = 1, \quad \gamma_{\mathrm{G}} = h/2. \tag{5.6}$$

Proof sketch. Test the equation $A_h^c V = d$ with V, bound the interior and boundary parts by the Cauchy–Schwarz inequality and use Lemma 5.2. The complete proof is given in Appendix A. □

Corollary 5.4 (Energy and discrete L^2 convergence)**.** *Under Assumption 2.1, the discretisation errors satisfy*

$$\|e_h^{\mathrm{FO}}\|_{2,h} \le \|e_h^{\mathrm{FO}}\|_{E,h,\mathrm{FO}} \le \tfrac{M_4}{6}h^2 + \tfrac{M_2}{2}h, \tag{5.7}$$

$$\|e_h^{\mathrm{G}}\|_{2,h} \le \|e_h^{\mathrm{G}}\|_{E,h,\mathrm{G}} \le \tfrac{M_4+M_3}{6}h^2 + \tfrac{M_4}{24}h^3. \tag{5.8}$$

Thus both norms have upper orders one (FO) and two (G).

Proof sketch. Apply Proposition 5.3 to the error equation with $F = -\tau_I$ and $S = -\tau_B^c$, bound the restricted norms by the maximum norms, and insert the defect bounds of Lemma 4.1; Lemma 5.2 gives the L^2 bound. The complete proof is given in Appendix A. □

The energy norm depends on the selected boundary closure. It provides a further theoretical and numerical check, while the common nodal maximum norm remains the primary target for comparing computational cost.

5.3 Solver implications of the proved matrix structure

The raw matrices are nonsymmetric and invertible. The direct baseline uses sparse LU factorisation. The supernodal approach underlying SuperLU is described by Demmel et al. [15]. The implementation calls SciPy's `spsolve` with CSC input and `use_umfpack=False` [6]. Each timed call includes a fresh factorisation.

Multiplication by the invertible positive diagonal matrix W_h^c gives the equivalent system

$$K_h^c U_h^c = W_h^c b_h^c, \qquad K_h^c = W_h^c A_h^c.$$

The weighted energy identity proves that K_h^c is symmetric positive definite. This establishes the matrix assumption required by the classical conjugate gradient method [13] and its SciPy implementation [7]. Moreover,

$$(K_h^c)_{kk} = e_k^T K_h^c e_k > 0,$$

so $\mathrm{diag}(K_h^c)^{-1}$ is a symmetric positive definite Jacobi preconditioner. CG is therefore applied to the weighted system.

For the raw nonsymmetric system, GMRES provides a Krylov method for nonsymmetric linear equations [14]. The ILU and Jacobi choices follow the standard preconditioning framework [16]. In SciPy's `gmres`, the parameter `M` applies the inverse preconditioner, while the final stopping test uses the original residual $b - AU$ [8]. The incomplete LU factors are supplied by `spilu` [9].

The numerical cross-checks record solver exit status and recompute the original-system residual. The structural analysis justifies the admissibility of these solver choices; their measured performance is assessed in the numerical experiments.

6 Verification

The conclusions of this report rest on a chain of evidence rather than on a single number. This section states which claims are made, which checks support each of them, and then reports the checks that concern the assembled equations and the linear solves. The convergence evidence itself is reported in Section 7.

6.1 Evidence plan

Table 2 lists each main claim together with the independent checks that support it. A check is counted as independent only if it does not share a derivation or an implementation with the code under test; agreement between two runs of the same code is not used as evidence.

Claim	Supporting evidence	Where
FO maximum error is $\Theta(h)$	Theorems 4.7 and 4.8; fitted orders on $N = 128, 256, 512$; FDM–FEM difference decays at first order	Sections 6.3 and 7.2
G error is $O(h^2)$	Theorem 4.7 and Corollary 5.4; fitted orders in the nodal maximum and discrete L^2 norms; FDM–FEM difference decays at second order	Sections 6.3 and 7.2
The assembled rows are the intended equations	Hand-derived boundary row; polynomial defect identities; separately assembled uneliminated system; independent P1 finite-element solve	Sections 6.2 and 6.3
Linear-solver error is negligible	Direct-solve residual and conditioning-based forward-error bound; ILU- and Jacobi-preconditioned GMRES tolerance sweeps with a residual rule; weighted CG	Section 6.4
The weighted matrices are symmetric positive definite	Proposition 5.1; edge-sum identity and smallest-eigenvalue estimate at $N = 32$	Sections 5 and 7.4
G reaches fixed accuracy targets on much smaller grids than FO	Timing protocol with warm-up and fixed error targets; equal-N comparison	Section 7.5

Table 2: Claims of this report and the checks that support them.

6.2 Equation-level checks

The following checks test whether the assembled matrix discretises the intended equations. None of them involves an iterative solver.

Hand-derived boundary row. At $N = 4$, $h = 1/4$, $j = 1$, the ghost row has diagonal $64+8\beta$, west coefficient -32 and retained north coefficient -16. After eliminating the bottom-right corner, its right-hand side is $(6 + 8\beta)e^{1.25} + 16e$. These values check the inward factor, outward-normal sign and corner contribution separately.

Manufactured functions with different polynomial structure. For $u = 1 + x + y$, both closures have zero defect in exact arithmetic. For $u = 1 + x^2 + y^2$, the ghost defect is zero and the FO boundary defect is $-h$. An additional check uses

$$u = 1 + x^3 + y^4, \qquad f = -6x - 12y^2, \qquad s_\beta = 3 + \beta(2 + y^4).$$

Direct polynomial substitution predicts

$$\tau_I = -2h^2, \qquad \tau_B^{\mathrm{FO}} = -3h + h^2, \qquad \tau_B^{\mathrm{G}} = 2h - 2h^2.$$

These identities are checked for both boundary variants at $N = 2, 3, 4, 6$, using data derived without applying the discrete matrix. They exercise nonzero forcing as well as boundary derivatives.

A separately assembled uneliminated system. On the same four small grids, an additional dense system retains every Dirichlet variable and, for G, every ghost variable. It imposes explicit Dirichlet identity equations, the original five-point PDE and centred boundary equations. Its physical-node solution is compared with the reduced sparse solve in all sixteen configurations. Agreement is required within absolute and relative tolerances 10^{-11}. This checks elimination and assembly through a different system layout.

Weighted-structure checks. For the energy identity, a seeded random vector is tested against the independent edge sums, with relative discrepancy below 10^{-12}. At $N = 32$, `eigsh` with `which='SA'` and tolerance 10^{-10} estimates the smallest eigenvalue [10]. This is a finite numerical corroboration of Proposition 5.1, not its proof. The results are reported in Section 7.4.

What these checks do not show. A dense solve of the same reduced matrix, or the iterative solves of Section 6.4, test linear algebra only. Agreement alone would not establish that the matrix discretises the intended PDE. The hand row, the independent data and the uneliminated construction supply the complementary equation checks, and the independent finite-element solve below supplies a check that shares neither derivation nor code with the finite-difference stencils.

6.3 Independent finite-element oracle

An independently assembled P_1 triangular finite-element solve uses the same continuum weak formulation: diffusion in Ω, Robin mass on Γ_B when $\beta = 2$, volume load f, boundary load s_β, and exact Dirichlet values on the other three edges. Continuous piecewise affine elements provide a conforming Galerkin discretisation in the standard finite-element framework [19]. The implementation uses scikit-fem [11], version 12.0.2 in the archived experiments, and does not call the finite-difference stencil assembly. The FEM and FDM solutions are compared at their common Cartesian vertices. These comparisons provide numerical evidence through a different discretisation of the same continuum problem.

Table 3 gives, for $N = 16, \ldots, 512$, the independent FEM error against e^{x+y} and the maximum difference between each FDM closure and the FEM solution at shared vertices. The FEM error converges at order 2.00. The FO–FEM difference converges at order 1.00 and the G–FEM difference at order 2.00, as expected when the FDM error is of the stated order and the FEM error is two orders smaller than the FO error. This agreement between two discretisations of the same weak problem is a cross-check, not a proof that either is exact.

Boundary	N	FEM error	p_{FEM}	FO–FEM	p	G–FEM	p
Neumann	16	2.29×10^{-3}	–	5.60×10^{-2}	–	3.15×10^{-3}	–
Neumann	32	5.74×10^{-4}	1.995	2.76×10^{-2}	1.023	7.90×10^{-4}	1.994
Neumann	64	1.44×10^{-4}	1.999	1.37×10^{-2}	1.011	1.98×10^{-4}	1.999
Neumann	128	3.59×10^{-5}	2.000	6.82×10^{-3}	1.006	4.94×10^{-5}	2.000
Neumann	256	8.98×10^{-6}	2.000	3.40×10^{-3}	1.003	1.24×10^{-5}	2.000
Neumann	512	2.24×10^{-6}	2.000	1.70×10^{-3}	1.001	3.09×10^{-6}	2.000
Robin	16	7.21×10^{-4}	–	3.36×10^{-2}	–	1.25×10^{-3}	–
Robin	32	1.81×10^{-4}	1.992	1.67×10^{-2}	1.010	3.14×10^{-4}	1.993
Robin	64	4.53×10^{-5}	1.999	8.31×10^{-3}	1.004	7.86×10^{-5}	1.999
Robin	128	1.13×10^{-5}	2.000	4.15×10^{-3}	1.003	1.97×10^{-5}	2.000
Robin	256	2.83×10^{-6}	2.000	2.07×10^{-3}	1.001	4.92×10^{-6}	2.000
Robin	512	7.08×10^{-7}	2.000	1.04×10^{-3}	1.001	1.23×10^{-6}	2.000

Table 3: Independent P1 triangular FEM solution of the weak form in Section 2, compared with both FDM closures at common grid vertices. FEM error: nodal maximum against the manufactured solution; FO–FEM and G–FEM: nodal maximum difference between the FDM and FEM solutions. Orders p are pairwise between adjacent grids; the differences are not error bounds.

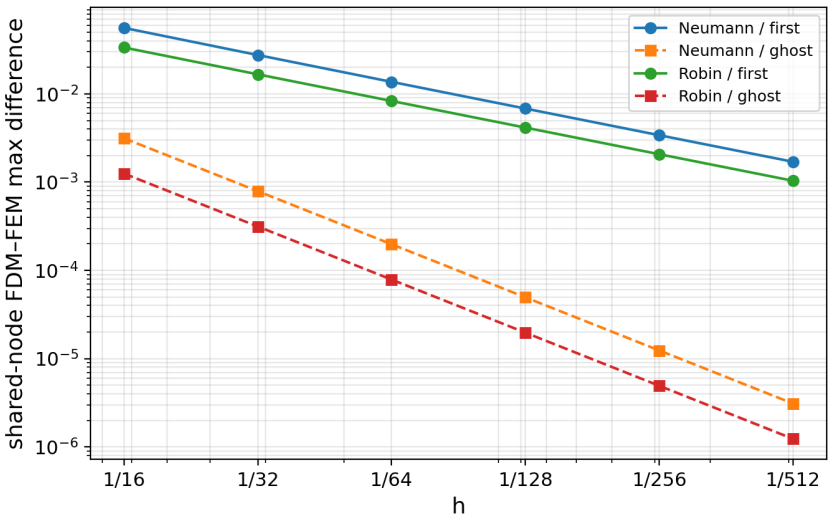


Figure 2: Maximum FDM–FEM difference at shared vertices. The FEM code assembles the continuum weak form independently of the FDM rows.

6.4 Separation of algebraic and discretisation error

Residual measures. For a computed vector $\widetilde{\boldsymbol{U}}$, set $\boldsymbol{r} = \boldsymbol{b} - A\widetilde{\boldsymbol{U}}$ and report

$$\eta_{\mathrm{cw}} = \max_k \frac{|r_k|}{(|A||\widetilde{\boldsymbol{U}}| + |\boldsymbol{b}|)_k}, \qquad \eta_{A,\infty} = \frac{\|\boldsymbol{r}\|_\infty}{\|A\|_\infty \|\widetilde{\boldsymbol{U}}\|_\infty + \|\boldsymbol{b}\|_\infty}. \tag{6.1}$$

A zero denominator with zero numerator contributes zero. The componentwise measure is unchanged by positive diagonal row scaling, so it compares the different raw row scales meaningfully. Neither residual is a discretisation error or a forward-error bound. The baseline is a sparse direct solve through SciPy's `spsolve` interface [6] with CSC input and `use_umfpack=False`, which fixes the SuperLU backend. Each call includes a fresh factorisation, and the solver has no user-set stopping tolerance. Its maximum componentwise residual over the twenty-four refinement runs is 2.33×10^{-12}.

Conditioning and forward error of the direct solve. Let $AU_h = b$ denote the exact discrete system, and let $\widetilde{U}$ be the computed direct solution. With

$$r = b - A\widetilde{U},$$

the exact-arithmetic identity $U_h - \widetilde{U} = A^{-1}r$ gives

$$\|U_h - \widetilde{U}\|_\infty \le \|A^{-1}\|_\infty \|r\|_\infty.$$

The distinction between residual measures, backward error and forward error is standard in numerical linear algebra [17].

For the conditioning diagnostic, the implementation estimates $\|A^{-1}\|_\infty = \|A^{-T}\|_1$ using the Higham–Tisseur block estimator [18], implemented by SciPy's `onenormest` [21]. The inverse actions are evaluated through the SuperLU factors. Define

$$\widehat{C} = \mathrm{onenormest}(A^{-T}), \qquad \widehat{\kappa}_\infty = \|A\|_\infty \widehat{C}, \qquad \widehat{B} = \widehat{C}\|r\|_\infty.$$

The routine supplies a lower estimate of the operator norm. Consequently, $\widehat{B}$ is a conditioning-based forward-error indicator, rather than a guaranteed upper bound.

A mesh-independent upper bound in exact arithmetic follows from Proposition 4.6. Apply that proposition to $e = U_h - \widetilde{U}$, with zero Dirichlet data and raw residual r. Writing $R = \|r\|_\infty$, its interior and boundary components are both bounded by R, and hence

$$\|U_h - \widetilde{U}\|_\infty \le B_{\mathrm{stab}}, \qquad B_{\mathrm{stab}} = \begin{cases} 7R, & c = \mathrm{FO}, \\ \dfrac{7}{4}(2+h)R, & c = \mathrm{G}. \end{cases}$$

Let

$$E_h = \|\widetilde{U} - u_{*,h}\|_\infty,$$

where $u_{*,h}$ contains the exact manufactured solution at the retained nodes. For the sixteen archived conditioning cases on $N = 16, 32, 64, 128$, substitution of the recorded residuals gives

$$\max \frac{B_{\mathrm{stab}}}{E_h} = 1.089 \times 10^{-4}.$$

Here E_h is the measured total error of the computed solution. The triangle inequality gives, whenever $B_{\mathrm{stab}} < E_h$,

$$\|U_h - u_{*,h}\|_\infty \ge E_h - B_{\mathrm{stab}},$$

and therefore

$$\frac{\|U_h - \widetilde{U}\|_\infty}{\|U_h - u_{*,h}\|_\infty} \le \frac{B_{\mathrm{stab}}}{E_h - B_{\mathrm{stab}}}.$$

The recorded values therefore support direct-solve errors well below one percent of the discretisation error on these sixteen cases.

The inequalities above are exact-arithmetic statements. The recorded residuals were evaluated in floating-point arithmetic without interval enclosures, so this assessment is not a formally rounded certificate. The archived conditioning study does not include $N = 256$ or 512.

Acceptance criterion. The iterative solves below are cross-checks on a representative mesh. Each is accepted when

$$\rho_{\mathrm{alg}} = \frac{\|U_{\mathrm{iterative}} - U_{\mathrm{direct}}\|_\infty}{\|U_{\mathrm{direct}} - u_h^*\|_\infty} < 10^{-2}. \tag{6.2}$$

The boundary-mean quantity $Q_h = (N-1)^{-1} \sum_{j=1}^{N-1} U_{N,j}$ is monitored as well. Agreement at tighter tolerances supports negligible solve error on this representative grid; it is not a general proof for arbitrary systems.

ILU-preconditioned GMRES. At $N = 64$, we cross-check the direct solution using GMRES [8]. Let $D_{kk} = (\sum_j |a_{kj}|)^{-1}$; GMRES solves $DAU = Db$ with ILU [9], drop tolerance 10^{-4}, fill factor 20, restart 50, at most 1000 restart cycles, zero initial vector, zero absolute tolerance and relative tolerances $10^{-8}, 10^{-10}, 10^{-12}$. The residual stopping test uses the scaled system's Euclidean norm; the original-system diagnostics in (6.1) are evaluated separately. D is not the analytical symmetriser W_h^c.

Table 4 implements this tolerance sweep. Every run returns status zero. Even at rtol 10^{-8} the largest ratio in (6.2) is 3.67×10^{-5}, below 10^{-2}; at 10^{-12} it is 4.34×10^{-8}. At the tightest tolerance, the largest change in the boundary-mean quantity relative to the direct solution is 2.24×10^{-12}. The residual, solution difference and quantity of interest thus provide separate evidence for the linear solve. Small nonmonotone ρ_{alg} changes between 10^{-10} and 10^{-12} correspond to absolute differences between 1×10^{-12} and 5×10^{-12} (the product of ρ_{alg} and the discretisation error). These small differences are consistent with floating-point effects in the direct and iterative solves. Their nonmonotonicity alone does not identify which solver contributes the dominant rounding error.

Boundary	Closure	rtol	Iterations	η_{cw}	ρ_{alg}
Neumann	FO	1×10^{-8}	4	5.92×10^{-11}	6.68×10^{-8}
Neumann	FO	1×10^{-10}	5	1.37×10^{-13}	3.31×10^{-10}
Neumann	FO	1×10^{-12}	5	9.57×10^{-14}	3.14×10^{-10}
Neumann	G	1×10^{-8}	4	7.09×10^{-11}	2.20×10^{-5}
Neumann	G	1×10^{-10}	5	1.95×10^{-13}	8.96×10^{-8}
Neumann	G	1×10^{-12}	5	6.52×10^{-14}	3.57×10^{-8}
Robin	FO	1×10^{-8}	4	6.48×10^{-11}	1.17×10^{-7}
Robin	FO	1×10^{-10}	5	2.14×10^{-13}	5.97×10^{-10}
Robin	FO	1×10^{-12}	5	1.20×10^{-13}	4.61×10^{-10}
Robin	G	1×10^{-8}	4	9.19×10^{-11}	3.67×10^{-5}
Robin	G	1×10^{-10}	5	1.21×10^{-13}	1.08×10^{-7}
Robin	G	1×10^{-12}	5	9.92×10^{-14}	4.34×10^{-8}

Table 4: GMRES–ILU cross-check at $N = 64$, with atol $= 0$. Residuals are evaluated on the original raw system. The ratio ρ_{alg} is the iterative/direct maximum difference divided by the measured direct-solution error; the acceptance target is 10^{-2}.

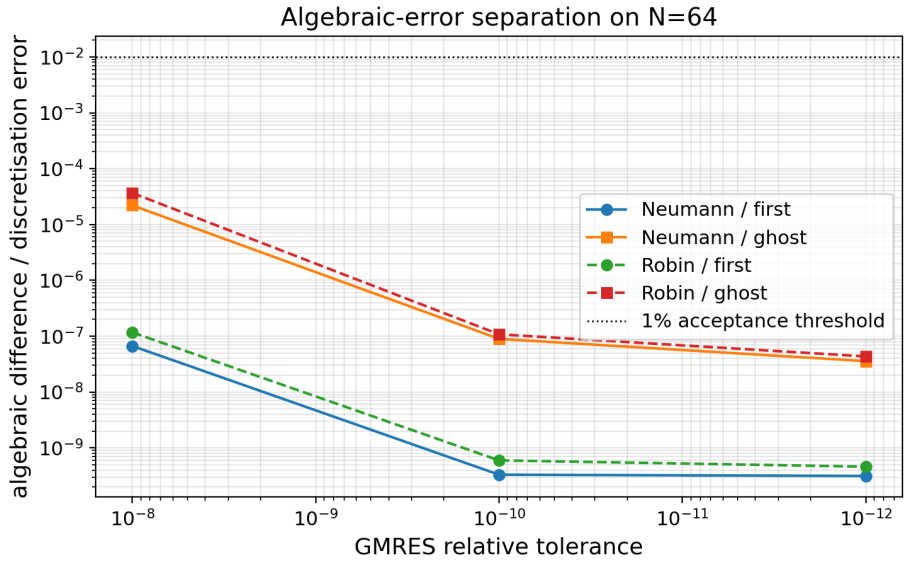


Figure 3: GMRES/direct differences relative to discretisation errors at $N = 64$. All ratios lie below the one-percent acceptance line. The flattening at tight tolerances is consistent with a solve-difference floor and is not interpreted as a new discretisation rate.

A weaker, genuinely iterative check and a stopping rule. ILU-preconditioned GMRES converges in only a few iterations and is a relatively weak independence check. Table 5 therefore uses GMRES on the original matrix with only a Jacobi (diagonal) preconditioner and no approximate factorisation: zero initial vector, zero absolute tolerance, restart 50, at most 1000 restart cycles and relative tolerances 10^{-8}, 10^{-10} and 10^{-12}. Iterations are counted from the residual-norm callback.

The bound of the previous paragraph also predicts how small the residual must be. If $\|\boldsymbol{r}\|_\infty / \|\boldsymbol{b}\|_\infty \le 10^{-2} E_{\infty,h} / (\|A^{-1}\|_\infty \|\boldsymbol{b}\|_\infty)$, then $\|U_{\mathrm{iterative}} - U_{\mathrm{direct}}\|_\infty \le 10^{-2} E_{\infty,h}$ up to the rounding error of the direct solve. This residual level is the column "Required" of Table 5. Since $\|A^{-1}\|_\infty$ is bounded and $\|\boldsymbol{b}\|_\infty$ grows like h^{-2}, the required relative residual shrinks like h^3 for FO and h^4 for G; for Neumann data it falls from 9.5×10^{-9} to 1.2×10^{-9} (FO) and from 1.5×10^{-10}

to 9.7×10^{-12} (G) between $N = 64$ and $N = 128$ (`data/condition_estimates.csv`). The rule is a sufficient condition, not a necessary one.

The prediction was checked against the runs. The rule is satisfied in ten of the twelve runs, and all ten meet the one-percent criterion. The two runs that violate it, the G closures at rtol 10^{-8}, are exactly the two that fail the criterion ($\rho_{\text{alg}} = 7.4 \times 10^{-2}$ for Neumann and 6.1×10^{-2} for Robin); tightening to 10^{-10} restores it. A stopping tolerance must therefore be chosen relative to the discretisation error of the closure being solved, and the same tolerance is stricter, in relative terms, for G than for FO.

Boundary	Closure	rtol	Iterations	ρ_{alg}	$\|r\|_\infty/\|b\|_\infty$	Required	Rule met
Neumann	FO	1×10^{-8}	452	3.14×10^{-4}	2.67×10^{-9}	9.46×10^{-9}	yes
Neumann	FO	1×10^{-10}	588	1.99×10^{-6}	4.04×10^{-11}	9.46×10^{-9}	yes
Neumann	FO	1×10^{-12}	718	2.18×10^{-8}	3.23×10^{-13}	9.46×10^{-9}	yes
Neumann	G	1×10^{-8}	435	7.40×10^{-2}	2.62×10^{-9}	1.52×10^{-10}	no
Neumann	G	1×10^{-10}	582	5.47×10^{-4}	3.25×10^{-11}	1.52×10^{-10}	yes
Neumann	G	1×10^{-12}	712	5.00×10^{-6}	2.89×10^{-13}	1.52×10^{-10}	yes
Robin	FO	1×10^{-8}	392	2.46×10^{-4}	2.72×10^{-9}	9.68×10^{-9}	yes
Robin	FO	1×10^{-10}	518	2.90×10^{-6}	3.83×10^{-11}	9.68×10^{-9}	yes
Robin	FO	1×10^{-12}	679	6.13×10^{-9}	6.90×10^{-14}	9.68×10^{-9}	yes
Robin	G	1×10^{-8}	393	6.14×10^{-2}	2.47×10^{-9}	1.10×10^{-10}	no
Robin	G	1×10^{-10}	519	7.47×10^{-4}	2.99×10^{-11}	1.10×10^{-10}	yes
Robin	G	1×10^{-12}	647	9.42×10^{-6}	2.30×10^{-13}	1.10×10^{-10}	yes

Table 5: Jacobi-preconditioned GMRES on the original matrix at $N = 64$, without ILU factorisation. "Required" is $10^{-2} E_{\infty,h}/(\|A^{-1}\|_\infty \|b\|_\infty)$, the relative residual that guarantees $\rho_{\text{alg}} < 10^{-2}$; the rule is sufficient, not necessary.

Weighted CG. Weighted CG uses $K_h^c U = W_h^c b$, Jacobi preconditioning, zero initial vector, rtol 10^{-10}, atol zero and at most 10000 iterations; CG applies because Proposition 5.1 proves the weighted matrix symmetric positive definite, and the Jacobi inverse $\text{diag}(K_h^c)^{-1}$ is an admissible positive definite preconditioner [7]. The reported normwise weighted residual is

$$\eta_{K,\infty} = \frac{\|Wb - K\widetilde{U}\|_\infty}{\|K\|_\infty \|\widetilde{U}\|_\infty + \|Wb\|_\infty}.$$

Table 6 also gives the original-system η_{cw}, so differently scaled residuals are not conflated. All four CG solves return status zero and require 280–289 iterations. The largest CG/direct difference ratio is 3.06×10^{-5}. This verifies the implementation of the proved eligible system. It does not establish that CG is faster than direct factorisation.

Boundary	Closure	Iterations	$\eta_{K,\infty}$	η_{cw}	ρ_{alg}
Neumann	FO	289	4.30×10^{-12}	1.99×10^{-11}	6.82×10^{-8}
Neumann	G	287	3.79×10^{-12}	1.66×10^{-11}	1.63×10^{-5}
Robin	FO	280	5.27×10^{-12}	2.57×10^{-11}	8.97×10^{-8}
Robin	G	280	3.84×10^{-12}	1.87×10^{-11}	3.06×10^{-5}

Table 6: Jacobi-preconditioned CG on $KU = Wb$ at $N = 64$, rtol 10^{-10}, atol zero. All return status zero. The two residual columns distinguish weighted normwise and original componentwise measures.

Scope. These comparisons use independent linear solvers for the same system; they do not replace the equation-level checks of Section 6.2. All iterative exit codes and residuals are recorded. The tolerance sweep is conducted on the required representative mesh rather than claimed as a forward-error proof for every possible refinement.

6.5 Reproducibility and test suite

The script `code/run_all.py` executes the four configurations, the six-mesh study, the fixed-target cost study and the solver cross-checks. It generates CSV data in `data/`, LaTeX tables in `generated/` and figures in `figures/`, so that no numerical value in the report is entered by hand. The assembly, solve and diagnostics live in `code/models.py`; the experiments in `code/experiments.py`; the independent finite-element solve in `code/fem_oracle.py`; and the pure-Neumann extension of Section 7.6 in `code/pure_neumann.py`. The pure-Neumann code treats all edges as flux boundaries, computes left- and right-null diagnostics and solves a bordered mean-zero system; it is not substituted for the mixed-boundary model. The script also estimates the conditioning of the direct solve `data/condition_estimates.csv`.

The executable test suite contains sixteen test methods. It checks dimensions, signs, finite entries, invalid options, corner ownership, exact rows, manufactured defects, observed rates, the weighted energy identity, the finite-element oracle, the pure-Neumann null space and gauge, the agreement of the ghost-closure compatibility defect with a trapezoidal quadrature, the one-norm estimate of $\|A^{-1}\|_\infty$ against a dense inverse, and its boundedness in h.

The bundled data were generated on Windows 11 with Python 3.13.5, NumPy 2.1.3, SciPy 1.15.3 and Matplotlib 3.10.0. The version and timing settings are recorded in `data/run_metadata.json`. Reproduction on another machine may change timings and the last few residual digits.

7 Numerical experiments

This section reports the main experiments. Sections 7.1 to 7.4 compare the two closures through convergence, error location and matrix structure; Section 7.5 compares their measured cost at fixed accuracy; Section 7.6 is a separate extension. The checks that establish that the numbers can be trusted are collected in Section 6.

7.1 Protocol, norms and order fitting

Both boundary variants and both closures use the same data, ordering and Dirichlet ownership. The required refinement family is $N = 16, 32, 64, 128$, extended to $N = 256, 512$ for finer-grid validation. For a computed full-grid error $e = \widetilde{U} - u_h^*$, we report $E_{2,h} = \|e\|_{2,h}$ and $E_{\infty,h} = \|e\|_{\infty,h}$ using exactly (3.3). At consecutive levels,

$$p_m(N) = \frac{\log(E_{m,2h}/E_{m,h})}{\log 2}, \qquad m = 2, \infty.$$

After inspecting pairwise trends, we fit $\log E = p \log h + C$ on $N = 128, 256, 512$. We also report the Richardson difference estimate $p_R = \log_2[(E_{128} - E_{256})/(E_{256} - E_{512})]$ when defined. Three-point fits alone do not prove an asymptotic expansion. Software versions, timing settings and the way the data are generated are described in Section 6.5.

7.2 Global convergence of the two closures

Boundary	Closure	N	h	$E_{2,h}$	p_2	$E_{\infty,h}$	p_∞	η_{cw}
Neumann	FO	16	1/16	1.786×10^{-2}	–	5.832×10^{-2}	–	1.94×10^{-14}
Neumann	FO	32	1/32	8.515×10^{-3}	1.069	2.815×10^{-2}	1.051	2.71×10^{-14}
Neumann	FO	64	1/64	4.161×10^{-3}	1.033	1.383×10^{-2}	1.025	1.10×10^{-13}
Neumann	FO	128	1/128	2.058×10^{-3}	1.016	6.853×10^{-3}	1.013	2.14×10^{-13}
Neumann	FO	256	1/256	1.023×10^{-3}	1.008	3.411×10^{-3}	1.007	5.94×10^{-13}
Neumann	FO	512	1/512	5.102×10^{-4}	1.004	1.702×10^{-3}	1.003	2.33×10^{-12}
Neumann	G	16	1/16	2.206×10^{-4}	–	8.611×10^{-4}	–	2.10×10^{-16}
Neumann	G	32	1/32	5.499×10^{-5}	2.004	2.163×10^{-4}	1.993	2.95×10^{-16}
Neumann	G	64	1/64	1.374×10^{-5}	2.001	5.412×10^{-5}	1.998	4.50×10^{-16}
Neumann	G	128	1/128	3.434×10^{-6}	2.000	1.354×10^{-5}	1.999	5.06×10^{-16}
Neumann	G	256	1/256	8.584×10^{-7}	2.000	3.385×10^{-6}	2.000	7.15×10^{-16}
Neumann	G	512	1/512	2.146×10^{-7}	2.000	8.462×10^{-7}	2.000	8.38×10^{-16}
Robin	FO	16	1/16	1.059×10^{-2}	–	3.430×10^{-2}	–	8.94×10^{-15}
Robin	FO	32	1/32	5.143×10^{-3}	1.042	1.685×10^{-2}	1.026	1.63×10^{-14}
Robin	FO	64	1/64	2.536×10^{-3}	1.020	8.355×10^{-3}	1.012	7.75×10^{-14}
Robin	FO	128	1/128	1.259×10^{-3}	1.010	4.159×10^{-3}	1.007	2.31×10^{-13}
Robin	FO	256	1/256	6.276×10^{-4}	1.005	2.075×10^{-3}	1.003	8.25×10^{-13}
Robin	FO	512	1/512	3.133×10^{-4}	1.002	1.036×10^{-3}	1.002	1.45×10^{-12}
Robin	G	16	1/16	1.273×10^{-4}	–	5.307×10^{-4}	–	3.31×10^{-16}
Robin	G	32	1/32	3.162×10^{-5}	2.009	1.332×10^{-4}	1.994	3.19×10^{-16}
Robin	G	64	1/64	7.892×10^{-6}	2.003	3.335×10^{-5}	1.998	4.52×10^{-16}
Robin	G	128	1/128	1.972×10^{-6}	2.001	8.340×10^{-6}	2.000	4.72×10^{-16}
Robin	G	256	1/256	4.930×10^{-7}	2.000	2.085×10^{-6}	2.000	7.82×10^{-16}
Robin	G	512	1/512	1.232×10^{-7}	2.000	5.213×10^{-7}	2.000	1.36×10^{-15}

Table 7: Global refinement errors for both boundary data and both closures; pairwise orders use adjacent grids.

Table 7 shows decreasing errors on every refinement. FO pairwise orders approach one from above in both global norms, while G orders are close to two. The maximum componentwise direct-solve residual across the twenty-four runs in Table 7 is 2.33×10^{-12}. Its small size is supplemented by the independent solve checks of Section 6.4; it is not used alone to establish correct boundary discretisation.

Boundary	Closure	p_2	p_∞	p_B	p_I	$p_{R,2}$	$p_{R,\infty}$
Neumann	FO	1.006	1.005	1.005	0.993	1.012	1.010
Neumann	G	2.000	2.000	2.000	1.984	2.000	2.000
Robin	FO	1.004	1.003	1.003	0.990	1.007	1.005
Robin	G	2.000	2.000	2.000	1.980	2.000	2.000

Table 8: Least-squares slopes and Richardson difference-based estimates on $N = 128, 256, 512$. Three points do not independently establish an asymptotic regime.

Table 8 distinguishes a theorem's upper order from a measured slope. The FO maximum-error rate is also supported by the independent lower bound in Theorem 4.8. For G, observed second order agrees with the proved upper bound; no mathematical lower bound for its error is inferred from these meshes.

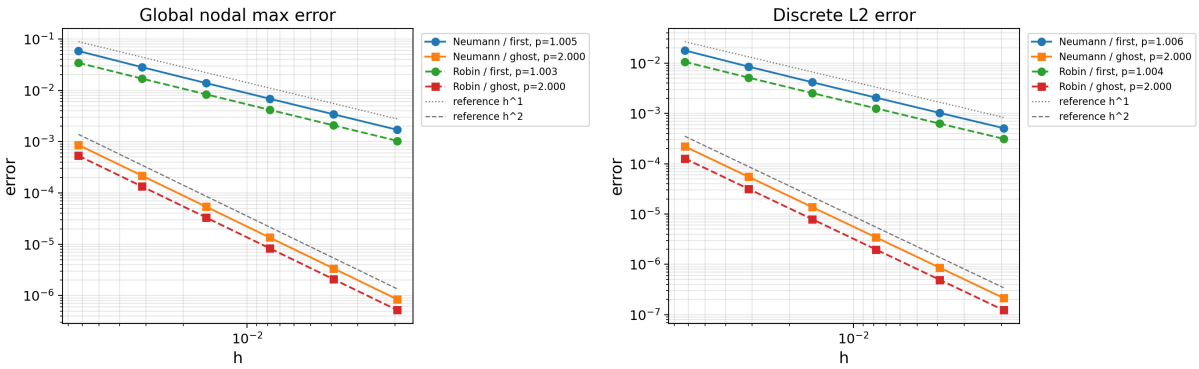


Figure 4: Global maximum and trapezoidal discrete L^2 refinement errors. Legends give the fitted slopes on $N = 128, 256, 512$; dotted/dashed reference curves have slopes one and two and arbitrary vertical placement. Errors approach the orders predicted by Theorem 4.7 and Corollary 5.4.

7.3 Boundary error, interior error and finite-grid deviations

We evaluate $E_{\infty,B} = \max_{1\le j<N} |e_{N,j}|$ and $E_{\infty,I} = \max_{(i,j)\in\mathcal{I}_h} |e_{i,j}|$ separately. The CSV data also retain the restricted L^2 errors defined by (3.4). This classifies errors by where they are measured. It is not an additive separation of the effects of boundary and interior truncation defects, which are coupled by the elliptic solve.

Boundary	Closure	N	$E_{\infty,B}$	p_B	$E_{\infty,I}$	p_I
Neumann	FO	16	5.832×10^{-2}	–	4.863×10^{-2}	–
Neumann	FO	32	2.815×10^{-2}	1.051	2.574×10^{-2}	0.918
Neumann	FO	64	1.383×10^{-2}	1.025	1.322×10^{-2}	0.961
Neumann	FO	128	6.853×10^{-3}	1.013	6.701×10^{-3}	0.981
Neumann	FO	256	3.411×10^{-3}	1.007	3.373×10^{-3}	0.990
Neumann	FO	512	1.702×10^{-3}	1.003	1.692×10^{-3}	0.995
Neumann	G	16	8.611×10^{-4}	–	6.706×10^{-4}	–
Neumann	G	32	2.163×10^{-4}	1.993	1.914×10^{-4}	1.809
Neumann	G	64	5.412×10^{-5}	1.998	5.096×10^{-5}	1.909
Neumann	G	128	1.354×10^{-5}	1.999	1.314×10^{-5}	1.956
Neumann	G	256	3.385×10^{-6}	2.000	3.334×10^{-6}	1.978
Neumann	G	512	8.462×10^{-7}	2.000	8.398×10^{-7}	1.989
Robin	FO	16	3.430×10^{-2}	–	2.854×10^{-2}	–
Robin	FO	32	1.685×10^{-2}	1.026	1.537×10^{-2}	0.893
Robin	FO	64	8.355×10^{-3}	1.012	7.978×10^{-3}	0.946
Robin	FO	128	4.159×10^{-3}	1.007	4.064×10^{-3}	0.973
Robin	FO	256	2.075×10^{-3}	1.003	2.051×10^{-3}	0.987
Robin	FO	512	1.036×10^{-3}	1.002	1.030×10^{-3}	0.993
Robin	G	16	5.307×10^{-4}	–	3.928×10^{-4}	–
Robin	G	32	1.332×10^{-4}	1.994	1.149×10^{-4}	1.774
Robin	G	64	3.335×10^{-5}	1.998	3.098×10^{-5}	1.890
Robin	G	128	8.340×10^{-6}	2.000	8.039×10^{-6}	1.946
Robin	G	256	2.085×10^{-6}	2.000	2.047×10^{-6}	1.973
Robin	G	512	5.213×10^{-7}	2.000	5.165×10^{-7}	1.987

Table 9: Spatially restricted maximum errors; this is not an additive source decomposition.

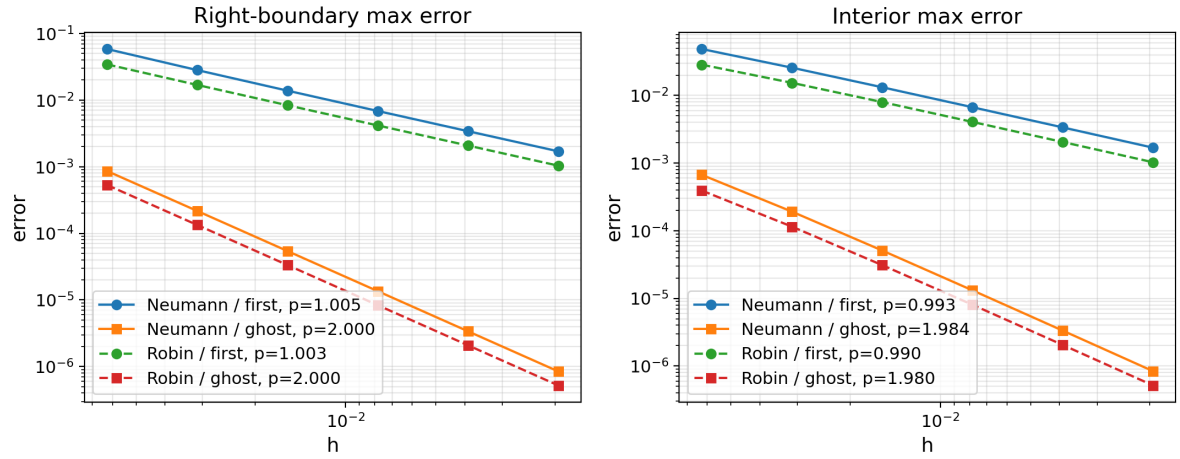


Figure 5: Error maxima on the open right edge and in the interior. The displayed slopes use the same finest-three-level fits. FO boundary error limits global accuracy, and the associated error extends into the interior. The spatial classification does not attribute interior error exclusively to the interior stencil.

For all tested configurations the global maximum equals the right-edge maximum, and the interior maximum is smaller but of comparable size. The interior FO fits approach one, consistent with a boundary defect affecting the full field. Thus a second-order interior stencil does not give a second-order global solution under the FO closure.

The G interior maximum fits are 1.984 and 1.980, slightly below the global fits of two. The pairwise orders in Table 9 increase monotonically towards two as the grid is refined, from about 1.8 at $N = 32$ to about 1.99 at $N = 512$ for both boundary variants. An additional location diagnostic, `data/error_locations.csv`, finds the fine-grid interior maxima on column $i = N-1$. This sampled column moves towards $x = 1$ as h decreases. The interior/global error ratios also move towards one. These observations indicate finite-grid and maximising-location effects; they do not identify a separate asymptotic leading term. The measured deviations are compatible with the uniform $O(h^2)$ upper bound and are not evidence of tolerance contamination, given the much smaller solver differences reported in Section 6.4.

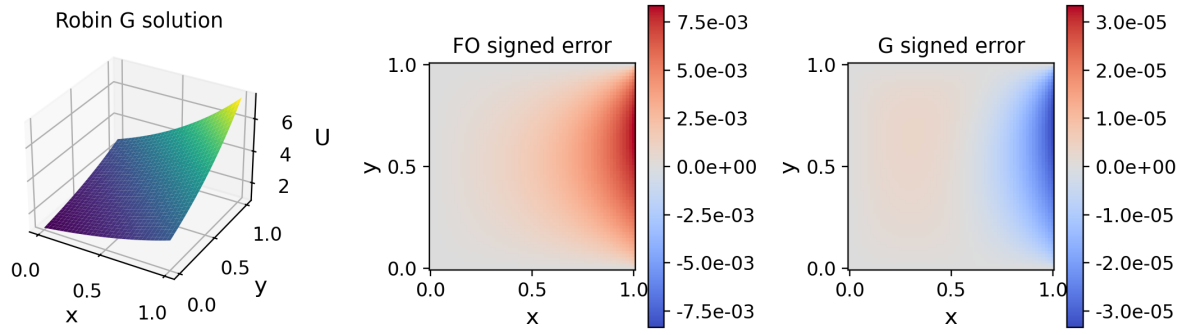


Figure 6: Representative Robin solution and signed errors at $N = 64$. The surface is the G solution. The FO and G error maps use separate, explicit colour scales centred at zero; their colours alone should not be compared as equal error magnitudes. All three Dirichlet edges and four corners have zero imposed-data error.

7.4 Matrix structure and weighted-energy norm

Raw matrices, weighted structure and sparsity. At $N = 32$, $n_{\mathrm{dof}} = 992$, and the exact counts in Proposition 3.2 give 4774 FO and 4834 G nonzeros. The spy plots show the extra right-edge tangential couplings in G. They do not themselves establish symmetry; the Frobenius diagnostic quantifies it.

Boundary	Closure	δ_A	δ_K	$\lambda_{\min}(K)$	Energy defect
Neumann	FO	0.05515	0	11.6098	1.17×10^{-16}
Neumann	G	0.05595	0	11.9409	0
Robin	FO	0.05515	0	14.7168	0
Robin	G	0.05590	0	14.8745	2.33×10^{-16}

Table 10: Raw and weighted matrix checks at $N = 32$: $\delta_A = \|A - A^T\|_F / \|A\|_F$ and $\delta_K = \|K - K^T\|_F / \|K\|_F$. Eigenvalues are numerical estimates; positive definiteness for all stated grids is proved in Proposition 5.1.

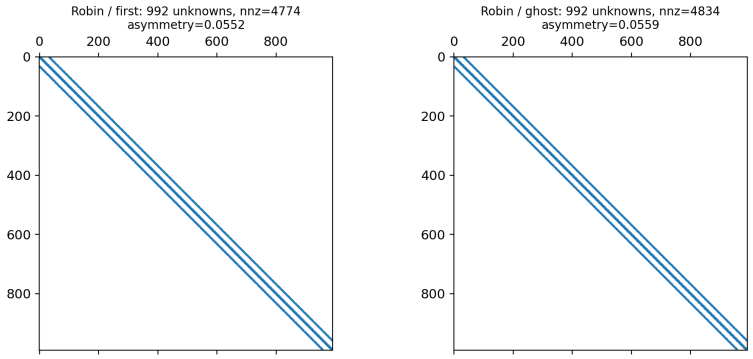


Figure 7: Representative Robin raw matrices at $N = 32$. The sparsity patterns retain nearest-neighbour couplings, while ghost elimination changes the inward coefficient and adds tangential boundary couplings. Both raw matrices are nonsymmetric, as confirmed by the labelled Frobenius ratios.

Table 10 gives nonzero raw asymmetry for all four configurations, exactly zero measured weighted asymmetry on this binary grid, and positive smallest-eigenvalue estimates. The edge-sum identity agrees with the quadratic form at roundoff. These checks corroborate the particular implementation of W_h^c, while Proposition 5.1 supplies the general proof.

Weighted-energy convergence. For the error restricted to unknowns we compute

$$E_{E,h,c} = \left(h^2 \boldsymbol{e}^T K_h^c \boldsymbol{e}\right)^{1/2}.$$

Each quadratic form is independently evaluated through the edge sums. The largest relative discrepancy over the refinement runs is 1.14×10^{-15}. The energy data for every grid used are stored in `data/energy_convergence.csv`.

Boundary	Closure	$E_{E,h}$ at $N = 512$	Fitted p_E	R_E^2
Neumann	FO	2.386×10^{-3}	1.003	1.000000
Neumann	G	1.262×10^{-6}	2.000	1.000000
Robin	FO	1.894×10^{-3}	1.002	1.000000
Robin	G	1.030×10^{-6}	2.000	1.000000

Table 11: Measured energy errors and fine-mesh fits on $N = 128, 256, 512$. The energy norm is closure dependent; it is not the shared cost-comparison norm.

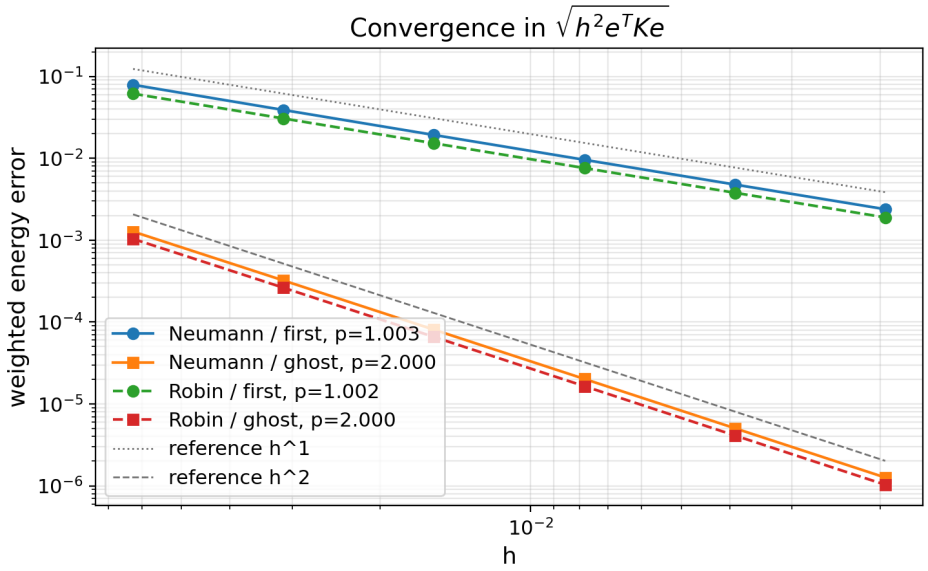


Figure 8: Convergence in the proved closure-dependent energy norm. Reference slopes one and two and fitted slopes on $N = 128, 256, 512$ are shown. FO and G approach the respective upper orders in Corollary 5.4.

The energy fits approach one and two, respectively. This additional norm checks the theory beyond nodal errors, but the two closures use different energy norms. The common nodal maximum error, rather than these closure-dependent energies, therefore defines the cost target in Section 7.5.

7.5 Measured cost at fixed accuracy thresholds

The cost measure is wall-clock time for assembly followed by a fresh sparse direct solve, including COO/CSR/CSC conversion, ordering, factorisation and triangular solves. Imports, a warm-up run, post-solve error/residual diagnostics, independent tests, Krylov and eigenvalue cross-checks, plotting and file I/O are excluded. After one warm-up, the reported cost is the median of the individually measured totals T_k obtained with `perf_counter`: 25 samples for $N \leq 32$, where runs last well under a millisecond, and 5 samples otherwise. Separate assembly and solve medians are displayed in Table 12, but their sum is not substituted for the independently measured total median. The sample minimum–maximum remains in the CSV. No factorisation or preconditioner is reused.

The fixed targets are $E_{\infty,h} \leq 5 \times 10^{-3}$, 2×10^{-3}, 10^{-3}, 10^{-4} and 10^{-5}, for each boundary variant separately. Candidate grids are $N \in \{4, 8, 16, 32, 64, 128, 256, 512\}$; the two extra coarse levels serve this cost study only and do not enter convergence fits. The selection rule chooses the coarsest tested grid meeting the target. If none does, the target is recorded as unattained; no

unmeasured runtime is extrapolated. This is a comparison at a common error bound, not at identical achieved error: G may oversatisfy a threshold, and the selected grid is not asserted to be a global runtime optimum.

Boundary	Target	Closure	N	$E_{\infty,h}$	Assembly (ms)	Solve (ms)	Total (ms)
Neumann	5×10^{-3}	FO	256	3.411×10^{-3}	14	280	300
Neumann	5×10^{-3}	G	8	3.377×10^{-3}	0.32	0.16	0.48
Neumann	2×10^{-3}	FO	512	1.702×10^{-3}	63	1900	2000
Neumann	2×10^{-3}	G	16	8.611×10^{-4}	0.47	0.58	1.1
Neumann	1×10^{-3}	FO	–	not reached	–	–	–
Neumann	1×10^{-3}	G	16	8.611×10^{-4}	0.47	0.58	1.1
Neumann	1×10^{-4}	FO	–	not reached	–	–	–
Neumann	1×10^{-4}	G	64	5.412×10^{-5}	0.89	8.3	9.2
Neumann	1×10^{-5}	FO	–	not reached	–	–	–
Neumann	1×10^{-5}	G	256	3.385×10^{-6}	19	300	320
Robin	5×10^{-3}	FO	128	4.159×10^{-3}	4.4	54	58
Robin	5×10^{-3}	G	8	2.066×10^{-3}	0.39	0.19	0.59
Robin	2×10^{-3}	FO	512	1.036×10^{-3}	70	1900	2000
Robin	2×10^{-3}	G	16	5.307×10^{-4}	0.49	0.61	1.1
Robin	1×10^{-3}	FO	–	not reached	–	–	–
Robin	1×10^{-3}	G	16	5.307×10^{-4}	0.49	0.61	1.1
Robin	1×10^{-4}	FO	–	not reached	–	–	–
Robin	1×10^{-4}	G	64	3.335×10^{-5}	1.1	9.9	11
Robin	1×10^{-5}	FO	–	not reached	–	–	–
Robin	1×10^{-5}	G	128	8.340×10^{-6}	3.8	53	57

Table 12: Coarsest successful grid in $\{4, 8, 16, 32, 64, 128, 256, 512\}$. Separate assembly, solve, and complete-total medians are shown; their medians need not sum. Each median follows one warm-up run and uses 25 samples for $N \leq 32$ and 5 samples otherwise. Small-grid timings are noise sensitive.

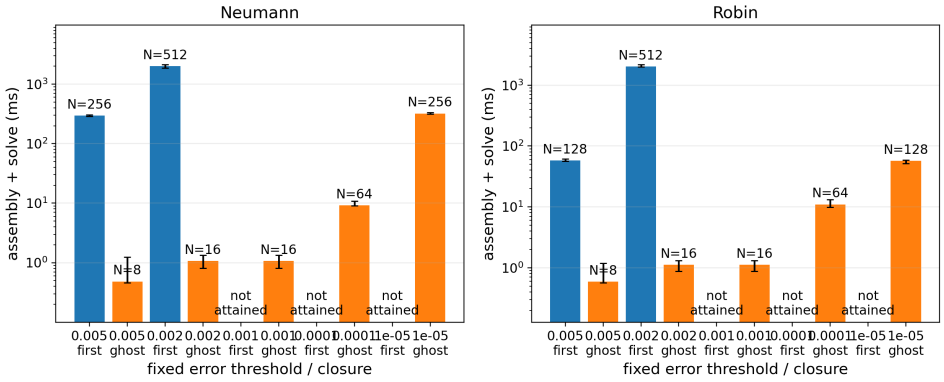


Figure 9: Measured cost under the same error thresholds within each boundary variant. The vertical axis is logarithmic. Error bars show the sample minimum–maximum; labels identify the selected grids. Only measured successful candidates are displayed.

Three observations follow. First, FO is first-order accurate, so within $N \leq 512$ it reaches only the two loosest targets; for 10^{-3} and below it is unattained, while G attains every target. Second, the advantage of G comes entirely from the coarser grid it needs, not from a cheaper solve per grid. At $N = 512$ the two matrices differ by only 0.08 % in nonzeros, and the measured totals are comparable (about 2.0 s for FO and 2.2 s for G with Neumann data, 2.0 s and 2.5 s with Robin data, `data/refinement_results.csv`). At equal N, G is therefore not cheaper, and was up to 1.2 times slower in this run. Third, the cost ratios at the two attained targets

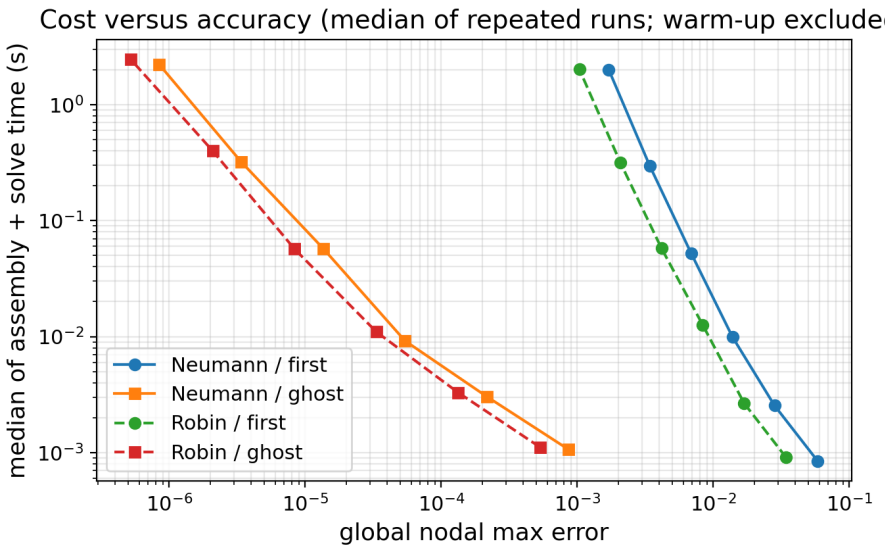


Figure 10: The full error–cost trajectories on the candidate grids $N = 4, \ldots, 512$. Cost is the median of measured assembly-plus-solve totals, including factorisation. This provides context; quantitative fixed-target comparisons use Table 12, not a visual comparison of unmatched points.

are large because they compare very different systems: $N = 256$ against $N = 8$ (56 unknowns, about 0.5 ms, close to timer resolution) and $N = 512$ (261,632 unknowns) against $N = 16$ (240 unknowns). The ratios are therefore evidence of the grid-size effect rather than precise speed-ups. Timings belong to this machine and run; ratios should not be transplanted to another machine, and no direct/CG runtime crossover is claimed from the single-grid CG cross-check.

If FO keeps its first-order law, $E_{\infty,h} \approx 0.87/N$ (Neumann) and $0.53/N$ (Robin), fitted from the $N = 512$ values, it would need roughly $N \approx 870$ and 530 for 10^{-3}, and $N \approx 8.7 \times 10^3$ and 5.3×10^3 for 10^{-4}, that is 7.6×10^7 and 2.8×10^7 unknowns. These figures are extrapolations, not measurements, and enter no cost ratio.

7.6 Extension: the pure-Neumann problem

This additional experiment is *not* the mixed problem above. All four edges now carry the outward flux $q = \partial_n e^{x+y}$ and all $(N+1)^2$ vertices, including corners, are unknown. The continuous compatibility condition is $\int_\Omega f + \int_{\partial\Omega} q = 0$. The FO scheme uses one-sided normal derivatives at edge nodes and the sum of the two outward derivative rows at corners. The G scheme uses centred ghost elimination in boundary PDE rows, adding both normal flux contributions at corners. Thus the finite-grid compatibility conditions need not agree.

For each singular matrix we compute a unit-sum left-null vector w and its defect $w^T b$. The constant right-null vector is tested through $\|A\mathbf{1}\|_\infty$. We solve the bordered system

$$\begin{bmatrix} A & \mathbf{1} \\ \mathbf{1}^T & 0 \end{bmatrix} \begin{bmatrix} U \\ \lambda \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix},$$

so U has arithmetic mean zero and $\lambda = w^T b$ is the compatibility correction. This adds the same scalar to every equation, although the rows of A have different scales. To test whether this choice drives the error, we also remove the defect with a correction proportional to $|\operatorname{diag} A|$, so that each row is perturbed in proportion to its own scale ("row-scaled" in Table 13). Errors compare U with the exact nodal values after the same mean-zero shift.

Table 13 and Figure 11 show three results. First, the FO defect and gauge error decrease approximately linearly in h, whereas G decreases approximately quadratically. Second, the

row-scaled correction does *not* reduce the FO error; at $N = 128$ it is 1.5×10^{-2} against 1.2×10^{-2} for the uniform correction. The $O(h)$ gauge error of FO is not explained by where the scalar correction is applied, and is consistent with the O(h) defect of its derivative rows. For G all rows have the same scale and the two corrections coincide. Third, the G compatibility defect equals the trapezoidal quadrature of the continuous compatibility integral to 10^{-13} or better, so it is the quadrature error of the data and not an assembly error. The diagnostics are saved in `data/pure_neumann_depth.csv`. The nonzero finite-grid defect is not concealed by the gauge correction.

Closure	N	$\|A\mathbf{1}\|_\infty$	Defect $w^T b$	Trapezoid defect	Gauge error (uniform)	Gauge error (row-scaled)	Residual
FO	8	0	1.61×10^{-1}	-7.70×10^{-3}	1.86×10^{-1}	2.22×10^{-1}	1.4×10^{-13}
FO	16	0	8.01×10^{-2}	-1.92×10^{-3}	9.45×10^{-2}	1.17×10^{-1}	4.1×10^{-13}
FO	32	0	4.00×10^{-2}	-4.81×10^{-4}	4.76×10^{-2}	5.98×10^{-2}	4.5×10^{-12}
FO	64	0	2.00×10^{-2}	-1.20×10^{-4}	2.39×10^{-2}	3.03×10^{-2}	2.8×10^{-11}
FO	128	0	9.99×10^{-3}	-3.00×10^{-5}	1.20×10^{-2}	1.52×10^{-2}	2.0×10^{-10}
G	8	0	-7.70×10^{-3}	-7.70×10^{-3}	1.21×10^{-2}	1.21×10^{-2}	3.4×10^{-13}
G	16	0	-1.92×10^{-3}	-1.92×10^{-3}	3.05×10^{-3}	3.05×10^{-3}	8.0×10^{-13}
G	32	0	-4.81×10^{-4}	-4.81×10^{-4}	7.65×10^{-4}	7.65×10^{-4}	5.1×10^{-12}
G	64	0	-1.20×10^{-4}	-1.20×10^{-4}	1.92×10^{-4}	1.92×10^{-4}	2.4×10^{-11}
G	128	0	-3.00×10^{-5}	-3.00×10^{-5}	4.79×10^{-5}	4.79×10^{-5}	1.2×10^{-10}

Table 13: Pure-Neumann extension with all boundary vertices unknown. The defect uses a unit-sum left null vector w; "Trapezoid defect" is the trapezoidal quadrature of the continuous compatibility integral. "Uniform" adds one scalar correction to every row (bordered solve); "row-scaled" removes the same defect with a correction proportional to $|\mathrm{diag}\, A|$. Both fix the arithmetic mean to zero. This is a distinct depth experiment, not the mixed problem's matrix.

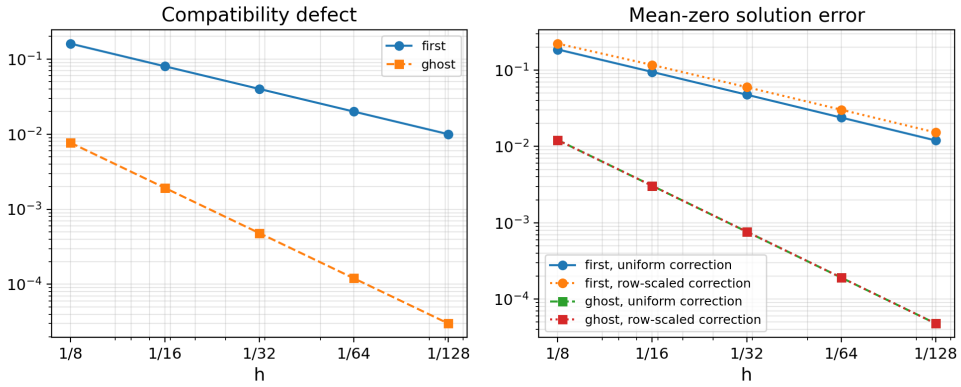


Figure 11: Pure-Neumann finite-grid compatibility defects and mean-zero solution errors for two distinct boundary closures.

8 Engineering interpretation and accuracy budgeting

The preceding analysis establishes convergence for smooth, consistent data, while the experiments compare the cost of reaching prescribed nodal-error thresholds. We now connect these results to two practical questions: how a temperature tolerance translates into a numerical accuracy requirement, and how errors in the prescribed data affect the discrete solution. This section derives consequences of the established stability analysis; it does not report an additional engineering validation experiment.

8.1 From a temperature tolerance to a grid choice

Under the scaling introduced in Section 2.1,

$$T(X) = T_{\mathrm{ref}} + \Delta T\, u(X/L), \qquad \Delta T > 0.$$

Consequently, a nodal approximation U_h satisfies

$$\max_{(x_i,y_j)} \left|T_{\mathrm{ref}} + \Delta T\, U_{i,j} - T\big(L(x_i,y_j)\big)\right| = \Delta T\, \|U_h - u_h\|_{\infty,h},$$

where u_h denotes the exact solution sampled at the grid nodes. A dimensional nodal-temperature tolerance $\varepsilon_T > 0$ therefore corresponds to the dimensionless requirement

$$\|U_h - u_h\|_{\infty,h} \le \frac{\varepsilon_T}{\Delta T}.$$

This conversion concerns nodal temperature values. It does not supply an error bound for a reconstructed temperature field or for a numerically differentiated heat flux.

For example, if $\varepsilon_T/\Delta T = 10^{-4}$, the archived fixed-accuracy study in Table 12 selects $N = 64$ for the ghost closure in both boundary variants. The first-order closure does not attain this threshold among the tested grids with $N \le 512$. This gives a concrete interpretation of the cost comparison: the higher-order boundary formulation reduces the grid resolution needed to meet the same temperature-error requirement.

The selected grid is the coarsest successful member of the tested candidate set. It is not a certified optimum over all possible grids. Moreover, the selection uses errors known from a manufactured solution; an application without an exact solution would require an independently justified error estimator or refinement procedure.

8.2 Sensitivity to errors in the prescribed data

A fine grid cannot by itself establish that the prescribed forcing and boundary data accurately describe a physical system. To examine this distinction, keep the grid, the corner convention and the coefficient $\beta \ge 0$ fixed. Let U_h^c and $\widehat{U}_h^c$ be the exact discrete solutions for data (f, g, s_β) and $(\widehat{f}, \widehat{g}, \widehat{s}_\beta)$, respectively, using the same closure $c \in \{\mathrm{FO}, \mathrm{G}\}$. Both vectors include their prescribed Dirichlet values.

Define the sampled data discrepancies by

$$\delta_f = \max_{I_h \cup B_h} |\widehat{f} - f|,$$
$$\delta_D = \max_{D_h} |\widehat{g} - g|,$$
$$\delta_s = \max_{B_h} |\widehat{s}_\beta - s_\beta|.$$

The maximum defining δ_f includes the right boundary because the ghost closure retains a boundary PDE row.

Proposition 8.1 (Discrete sensitivity to prescribed data)**.** *For $N \ge 2$ and fixed $\beta \ge 0$,*

$$\|\widehat{U}_h^{\mathrm{FO}} - U_h^{\mathrm{FO}}\|_{\infty,h} \le \delta_D + \frac{7}{2}(\delta_f + \delta_s), \tag{8.1}$$

$$\|\widehat{U}_h^{\mathrm{G}} - U_h^{\mathrm{G}}\|_{\infty,h} \le \delta_D + \frac{7}{4}((2+h)\delta_f + 2\delta_s). \tag{8.2}$$

These estimates concern perturbations of the data at a fixed coefficient β.

Proof. Write $V = \widehat{U}_h^c - U_h^c$ and use the full-grid operators before Dirichlet elimination. Then

$$|V| \le \delta_D \quad \text{on } D_h, \qquad |L_h V| \le \delta_f \quad \text{on } I_h.$$

For the first-order closure,

$$|B_h^\beta V| \le \delta_s \quad \text{on } B_h.$$

For the ghost closure, subtraction of the two boundary PDE rows gives

$$|G_h^\beta V| \le \delta_f + \frac{2\delta_s}{h} \quad \text{on } B_h.$$

Use the barrier

$$\psi(x) = 1 + x - \frac{x^2}{4}.$$

The identities established in Section 4 give

$$1 \le \psi \le \frac{7}{4}, \qquad L_h\psi = \frac{1}{2}, \qquad B_h^\beta\psi \ge \frac{1}{2}, \qquad G_h^\beta\psi \ge \frac{1}{h}.$$

Set

$$Z = \delta_D + a\psi, \qquad a = \begin{cases} 2(\delta_f + \delta_s), & c = \text{FO}, \\ (2+h)\delta_f + 2\delta_s, & c = \text{G}. \end{cases}$$

The constant contribution δ_D has zero interior operator and nonnegative boundary operator, since $\beta \ge 0$. Therefore $Z \pm V$ is nonnegative on D_h and has nonnegative interior and boundary rows. For the ghost boundary, in particular,

$$\frac{a}{h} = \left(\frac{2}{h} + 1\right)\delta_f + \frac{2\delta_s}{h} \ge \delta_f + \frac{2\delta_s}{h}.$$

The discrete comparison principle yields $-Z \le V \le Z$. Taking the maximum and using $\psi \le 7/4$ proves both estimates. □

An important feature of (8.2) is that a boundary-data discrepancy δ_s contributes a mesh-independent term. The factor $2/h$ in the ghost boundary load is compensated by the factor h in its boundary stability estimate. Thus the raw load scaling does not imply an h^{-1} amplification of boundary-data errors in the solution.

For the convective interpretation in Section 2.1, $s_\beta = \beta u_\infty$. If β is fixed and only the dimensionless ambient-temperature trace is perturbed, then

$$\delta_s \le \beta \|\widehat{u}_\infty - u_\infty\|_{\infty,\Gamma_B}.$$

Both estimates consequently bound the associated solution change by a contribution no larger than

$$\frac{7}{2}\,\beta \|\widehat{u}_\infty - u_\infty\|_{\infty,\Gamma_B}.$$

This is an upper sensitivity estimate. It does not prove that every nonzero data perturbation produces a nonzero error of this size, or establish a positive lower error floor.

8.3 Separating the accuracy requirements

Let $\widetilde{U}_h^c$ be a computed approximation to the perturbed discrete solution $\widehat{U}_h^c$, extended by the same perturbed Dirichlet values. The triangle inequality gives

$$\begin{aligned} \|\widetilde{U}_h^c - u_h\|_{\infty,h} \le{}& \|\widetilde{U}_h^c - \widehat{U}_h^c\|_{\infty,h} + \|\widehat{U}_h^c - U_h^c\|_{\infty,h} \\ &+ \|U_h^c - u_h\|_{\infty,h}. \end{aligned} \tag{8.3}$$

The three terms are, respectively, the algebraic solve error, the effect of prescribed-data discrepancies, and the discretisation error for the nominal problem.

Section 6.4 supplies residual-based estimates for the first term, subject to its stated floating-point qualifications. Proposition 8.1 bounds the second term. The convergence estimates in Section 4 bound the third term under the smoothness and consistency assumptions of the nominal problem. Together they provide an accuracy budget rather than a single undifferentiated error measure.

For a target $\varepsilon_T/\Delta T$, one can assign nonnegative allowances to these three contributions whose sum does not exceed the target. Mesh refinement addresses the discretisation allowance; a tighter linear solve addresses the algebraic allowance; improved measurements or better prescribed data address the data allowance. The data-sensitivity bounds contain terms that do not decrease with h, so mesh refinement alone cannot reduce every contribution to the guaranteed total bound.

This interpretation explains both the value and the limits of the ghost closure. Its second-order convergence reduces the cost of controlling discretisation error in the smooth benchmark. It does not establish the accuracy of physical inputs or account for model-form errors, uncertain conductivity, an uncertain Robin coefficient, or unresolved geometric features. Those effects lie outside the fixed-operator perturbation estimate above and would require additional analysis or validation.

9 Conclusions and limitations

The right-edge equation controls the accuracy retained by the mixed Poisson discretisation. The two raw boundary defects are $O(h)$, but the comparison estimate weights the G defect by h. Together with the $O(h^2)$ interior defect, this yields the FO $O(h)$ and G $O(h^2)$ maximum-norm bounds. For e^{x+y}, the separate FO lower bound proves that first order is sharp in the boundary and global maximum norms. The weighted energy identity supplies a second stability route and discrete L^2/energy bounds for the actual boundary and corner layout.

Six-mesh experiments for both Neumann and Robin data reproduce the predicted approximately first- and second-order behaviour. The interior maximum-error fits exhibit visible finite-grid deviations that are reported rather than rounded to the theoretical orders. Independent boundary rows, different manufactured functions and an uneliminated system check test the equations. Direct/GMRES comparisons and weighted CG show solve differences much smaller than the measured discretisation error on the representative grid. An independent P1 FEM solve confirms the mixed weak-form benchmark at shared nodes. A separate pure-Neumann experiment exposes the different finite-grid compatibility defects and the constant nullspace, with a mean-zero gauge. The implemented positive row weights recover the proved symmetric positive definite structure without asserting symmetry of the raw systems.

At $h = 1/512$ the nodal maximum errors are 1.702×10^{-3} (FO) and 8.462×10^{-7} (G) for Neumann data, and 1.036×10^{-3} and 5.213×10^{-7} for Robin data. At equal N the two closures cost about the same, since the matrices differ by 0.08 % in nonzeros. At fixed nodal-error thresholds, G is therefore cheaper only because it meets each threshold on a much coarser grid: it attains all five thresholds from 5×10^{-3} to 10^{-5}, while FO attains only the two loosest within $N \leq 512$. The finite candidate set, the coarsest-successful-grid policy and the timing variability limit the scope of this efficiency conclusion.

The mathematical assumptions remain essential: bounded derivatives, fixed nonnegative β, exact imposed Dirichlet values and the stated square geometry. Smoothness is satisfied by the manufactured function and is not proved for arbitrary mixed data or corner-singular problems. The G theorem is an upper bound without a matching lower bound. Floating-point diagnostics and representative solver cross-checks are evidence for the reported computations, not universal forward-error certificates. The energy norms depend on the closure; common nodal errors

define the cost targets. Finally, this is verification on known solutions rather than experimental validation of a particular device.

The analysis and computations jointly support the conclusion that the ghost-eliminated formulation preserves second-order accuracy for this smooth benchmark, whereas the backward derivative formulation has a provably first-order maximum error. The distinction follows from the complete boundary formulation and its stability, not from the interior stencil alone.

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A Proofs of the analytical results

This appendix contains the complete proofs of the statements whose proof sketches appear in Sections 2–5. Statements and numbering are those of the main text; the proofs of Proposition 3.1, Corollary 4.2, Lemma 4.5, Proposition 4.6 and Theorem 4.7, which carry the main stability argument, remain in the main text.

Proofs for Section 2

Proof of Lemma 2.2. For almost every y, the one-dimensional H^1 slice has an absolutely continuous representative and its left trace is zero. Hence $v(x,y) = \int_0^x v_x(t,y)\,dt$. Cauchy–Schwarz gives

$$|v(x,y)|^2 \le x \int_0^x |v_x(t,y)|^2\,dt.$$

Integrate first in x, interchange the nonnegative integrals, and then integrate in y:

$$\begin{aligned}\|v\|_{L^2(\Omega)}^2 &\le \int_0^1 \int_0^1 \left(\int_t^1 x\,dx\right) |v_x(t,y)|^2\,dt\,dy \\ &= \int_0^1 \int_0^1 \tfrac{1-t^2}{2} |v_x(t,y)|^2\,dt\,dy \le \tfrac{1}{2}\|v_x\|_{L^2(\Omega)}^2.\end{aligned}$$

At $x = 1$, the same representation yields $|v(1,y)|^2 \le \int_0^1 |v_x(t,y)|^2\,dt$, proving the trace bound. Finally $\|v\|_{H^1}^2 \le \frac{3}{2}\|\nabla v\|_{L^2}^2$, while $a_\beta(v,v) \ge \|\nabla v\|_{L^2}^2$ because $\beta \ge 0$. Combining these inequalities gives (2.9). □

Proof of Proposition 2.3. The assumed trace of u shows $u \in V_g$. Green's identity, applicable to this smooth u and $v \in H^1(\Omega)$, gives

$$\int_\Omega \nabla u \cdot \nabla v = \int_\Omega f v + \int_{\partial\Omega} \partial_n u\, v = \int_\Omega f v + \int_{\Gamma_B} (s_\beta - \beta u) v \qquad (v \in V_0).$$

The contribution from Γ_D vanishes because the trace of v is zero there. Moving the Robin term to the left proves (2.6) and establishes existence.

For uniqueness, let $u_1, u_2 \in V_g$ solve (2.6) and put $w = u_1 - u_2 \in V_0$. Subtraction gives $a_\beta(w,v) = 0$ for all $v \in V_0$. Taking $v = w$ yields

$$\|\nabla w\|_{L^2(\Omega)}^2 + \beta \|\operatorname{Tr} w\|_{L^2(\Gamma_B)}^2 = 0.$$

Lemma 2.2 gives $\frac{2}{3}\|w\|_{H^1(\Omega)}^2 \le a_\beta(w,w) = 0$. Therefore $w = 0$ in $H^1(\Omega)$, and $u_1 = u_2$. □

Proofs for Section 3

Proof of Proposition 3.2. Every interior row contains a diagonal and an east unknown, contributing $2(N-1)^2$ entries. West unknowns occur for $i \ge 2$, contributing $(N-2)(N-1)$ entries. Each of the two vertical directions contributes $(N-2)(N-1)$ entries. Thus the interior rows contain $2(N-1)^2 + 3(N-2)(N-1) = (N-1)(5N-8)$ entries. Every boundary row contains its diagonal and west unknown, adding $2(N-1)$ entries. The FO closure has no further entries, giving its count in (3.13). The G closure also has $N-2$ unknown neighbours in each vertical direction, giving the additional $2(N-2)$ entries. All counted off-diagonals are strictly negative and all diagonals are strictly positive; none vanish for the stated β. Dirichlet entries have already been removed. These counts include $N = 2$, when the additional vertical-boundary count is zero. Every unknown contributes a nonzero diagonal, giving $\operatorname{nnz}(A_h^c) \ge n_{\mathrm{dof}}$. The formulas also give $\operatorname{nnz}(A_h^c) \le 5n_{\mathrm{dof}}$ for both closures and all $N \ge 2$. These two bounds prove the stated linear growth. □

Proofs for Section 4

Proof of Lemma 4.1. Taylor's theorem along the x grid line gives, at an interior node,

$$u(x_i \pm h, y_j) = \sum_{\ell=0}^{5} \frac{(\pm h)^\ell}{\ell!} \partial_x^\ell u(x_i, y_j) + R_\pm, \qquad |R_\pm| \le C_6 h^6/720,$$

where C_6 is a uniform bound for the pure sixth derivatives on $\overline{\Omega}$. Adding the expansions cancels all odd terms and yields $D_{xx}^h u = u_{xx} + h^2 u_{xxxx}/12 + O(h^4)$. The same computation in y, followed by $f = -\Delta u$, proves (4.3). Using Taylor's theorem only through order three instead bounds each centred second-derivative defect by $M_4 h^2/12$, proving the first inequality in (4.6).

At a right-edge node, write $u = u(1, y_j)$ and $u_W = u(1-h, y_j)$. Taylor's theorem gives

$$u_W = u - h u_x + \frac{h^2}{2} u_{xx} - \frac{h^3}{6} u_{xxx} + O(h^4).$$

Substitution in $(u - u_W)/h + \beta u - (u_x + \beta u)$ proves (4.4). More directly, the second-order Taylor remainder has magnitude at most $M_2 h^2/2$, giving $|\tau_B^{\mathrm{FO}}| \le M_2 h/2$.

For the ghost row, expand the inward and tangential terms separately:

$$2u - 2u_W = 2hu_x - h^2 u_{xx} + \frac{h^3}{3} u_{xxx} - \frac{h^4}{12} u_{xxxx} + O(h^5),$$

$$2u - u(1, y_j - h) - u(1, y_j + h) = -h^2 u_{yy} - \frac{h^4}{12} u_{yyyy} + O(h^6).$$

After division by h^2 and addition of $2\beta u/h$, this becomes

$$G_h^\beta u_h = \frac{2}{h}(u_x + \beta u) - \Delta u + \frac{h}{3} u_{xxx} - \frac{h^2}{12}(u_{xxxx} + u_{yyyy}) + O(h^3).$$

The first two terms equal $2s_\beta/h + f$, proving (4.5). For its uniform bound, use the exact decomposition

$$\tau_B^{\mathrm{G}} = \left[\frac{2}{h^2}(u - u_W - h u_x) + u_{xx}\right] + (u_{yy} - D_{yy}^h u_h).$$

The third-order Taylor remainder for u_W bounds the bracket by $M_3 h/3$. The centred tangential defect is at most $M_4 h^2/12$. The triangle inequality proves the final bound. Every Taylor segment used here lies in $\overline{\Omega}$, including when $N = 2$; bounded derivatives therefore make all constants independent of the node and mesh. □

Proof of Lemma 4.3. Suppose instead that the minimum of V is $m < 0$. The finite grid contains a node attaining m, which cannot belong to $\mathcal{D}_h$. At an interior minimum $P = (i, j)$,

$$0 \le (L_h V)_P = \frac{1}{h^2} \sum_{Q \in \{(i-1,j),(i+1,j),(i,j-1),(i,j+1)\}} (m - V_Q) \le 0.$$

Each summand is nonpositive and their sum is zero, so every neighbour equals m. In particular, the west neighbour is another minimum. Repeat this argument with decreasing i until reaching $(0, j)$, contradicting the nonnegative Dirichlet value.

If the initial minimum is at (N, j), the FO row gives

$$0 \le (B_h^\beta V)_j = \frac{m - V_{N-1,j}}{h} + \beta m \le 0.$$

For $\beta > 0$ the final inequality is strict, an immediate contradiction. For $\beta = 0$, equality forces $V_{N-1,j} = m$; since $N \ge 2$, this is an interior node and the preceding argument applies. For the ghost row,

$$0 \le (G_h^\beta V)_j = \frac{2(m - V_{N-1,j}) + (m - V_{N,j-1}) + (m - V_{N,j+1})}{h^2} + \frac{2\beta}{h} m \le 0.$$

Again $\beta > 0$ gives a contradiction. If $\beta = 0$, every weighted nonpositive term must vanish; in particular $V_{N-1,j} = m$, so the same interior argument applies. These cases cover every possible minimum. □

Proof of Proposition 4.4. The sign of every off-diagonal entry follows directly from the stencils. If $A_h^c \boldsymbol{V} = 0$, extend $\boldsymbol{V}$ by zero on $\mathcal{D}_h$. Its full-grid interior and boundary rows are zero. Lemma 4.3 applied to both V and $-V$ gives $V = 0$. The square matrix A_h^c therefore has a trivial kernel and is invertible. For any vector $\boldsymbol{d} \geq 0$, its solution $\boldsymbol{V} = (A_h^c)^{-1}\boldsymbol{d}$, extended by zero, satisfies the hypotheses of that lemma and hence is nonnegative. Taking $\boldsymbol{d}$ to be each coordinate vector proves that every column of $(A_h^c)^{-1}$ is nonnegative. □

Proof of Theorem 4.8. For the exponential, the interior identity (4.7) and $\cosh h - 1 \geq h^2/2$ give

$$\frac{\tau_I}{u^*} = 2 - \frac{4(\cosh h - 1)}{h^2} \leq 0.$$

At a boundary node, cancellation of the Robin term and the fundamental theorem of calculus give the exact formula

$$-\tau_{B,j}^{\mathrm{FO}} = \frac{1}{h}\int_{1-h}^{1} (z - 1 + h) u_{xx}^*(z, y_j)\, dz.$$

Since $y_j \geq h$ and $z \geq 1 - h$, $u_{xx}^*(z, y_j) = e^{z+y_j} \geq e$. The integral of $z - 1 + h$ is $h^2/2$, so $-\tau_{B,j}^{\mathrm{FO}} \geq eh/2$.

Set $\phi_{i,j} = \sinh(\pi x_i)\sin(\pi y_j)$. It vanishes on $\mathcal{D}_h$ and is nonnegative on the full grid. Direct use of the addition formulas yields

$$(L_h\phi)_{i,j} = \frac{4 - 2\cosh(\pi h) - 2\cos(\pi h)}{h^2}\phi_{i,j} \leq 0.$$

The sign follows from $\cosh t + \cos t = 2\sum_{k=0}^{\infty} t^{4k}/(4k)! \geq 2$ for real t. On the right edge,

$$(B_h^\beta \phi)_j = \left[\frac{\sinh \pi - \sinh(\pi(1-h))}{h} + \beta \sinh \pi\right] \sin(\pi y_j) \leq m_\beta.$$

Indeed the backward quotient equals the average of $\pi\cosh(\pi x)$ over $[1 - h, 1]$, which is at most $\pi \cosh \pi$, and $0 \leq \sin(\pi y_j) \leq 1$.

Choose $a = e/(2m_\beta) > 0$ and define $V = e_h^{\mathrm{FO}} - ah\phi$. It has zero Dirichlet values. The error equation gives

$$L_h V = -\tau_I - ahL_h\phi \geq 0, \qquad B_h^\beta V = -\tau_B^{\mathrm{FO}} - ahB_h^\beta\phi \geq eh/2 - ahm_\beta = 0.$$

Lemma 4.3 implies $e_h^{\mathrm{FO}} \geq ah\phi$. Take a boundary index j closest to $N/2$. It belongs to $\{1, \ldots, N-1\}$ and satisfies $|y_j - 1/2| \leq h/2 \leq 1/4$, so $\sin(\pi y_j) \geq 1/\sqrt{2}$. Therefore $e_{N,j}^{\mathrm{FO}} \geq ah\sinh \pi/\sqrt{2} = C_\beta h$. The maximum-norm upper bound in Theorem 4.7 also bounds the boundary maximum by a constant times h. These two-sided bounds prove both $\Theta(h)$ statements. □

Proofs for Section 5

Proof of Proposition 5.1. Interior–interior off-diagonal pairs already agree. On each interior–boundary pair, multiplying the boundary coefficient by $1/h$ (FO) or $1/2$ (G) makes both coefficients $-1/h^2$. In G the two tangential boundary coefficients both become $-1/(2h^2)$; in FO both are zero. There are no other off-diagonal pairs, so K_h^c is symmetric.

For a fixed j, abbreviate $v_i = V_{i,j}$, with $v_0 = 0$. Expansion of the squares proves the finite summation identity

$$\sum_{i=1}^{N-1} v_i(2v_i - v_{i-1} - v_{i+1}) + v_N(v_N - v_{N-1}) = \sum_{i=0}^{N-1}(v_{i+1} - v_i)^2. \tag{A.1}$$

Indeed both sides equal $2\sum_{i=1}^{N-1} v_i^2 + v_N^2 - 2\sum_{i=1}^{N-1} v_i v_{i+1}$. For a vertical sequence z_j with $z_0 = z_N = 0$, the same expansion gives

$$\sum_{j=1}^{N-1} z_j(2z_j - z_{j-1} - z_{j+1}) = \sum_{j=0}^{N-1} (z_{j+1} - z_j)^2. \tag{A.2}$$

Multiplication of the scaled matrix by h^2 leaves the usual interior rows and the following boundary rows:

$$\begin{cases} V_{N,j} - V_{N-1,j} + \beta h V_{N,j}, & c = \mathrm{FO}, \\ V_{N,j} - V_{N-1,j} + \frac{1}{2}(2V_{N,j} - V_{N,j-1} - V_{N,j+1}) + \beta h V_{N,j}, & c = \mathrm{G}. \end{cases}$$

Apply (A.1) to every horizontal sequence and (A.2) to each interior column. In G also apply the vertical identity to the right column with factor $1/2$. Summing the remaining diagonal Robin terms proves (5.3), including the boundary edges adjoining the Dirichlet corners.

Every term on the right of (5.3) is nonnegative. If the quadratic form is zero, $S_x(V) = 0$, so every horizontal difference is zero. Starting from $V_{0,j} = 0$ then gives $V_{i,j} = 0$ for all unknowns. The symmetric quadratic form is therefore strictly positive for every nonzero vector, proving positive definiteness. □

Proof of Lemma 5.2. For fixed j, put $d_{r,j} = V_{r+1,j} - V_{r,j}$. Since $V_{0,j} = 0$, $V_{i,j} = \sum_{r=0}^{i-1} d_{r,j}$. The Cauchy–Schwarz inequality gives

$$|V_{i,j}|^2 \le i \sum_{r=0}^{i-1} d_{r,j}^2 \le N \sum_{r=0}^{N-1} d_{r,j}^2.$$

Using the zero Dirichlet values in (3.3), and $hN = 1$,

$$\|V\|_{2,h}^2 = h^2 \sum_{j=1}^{N-1} \left(\sum_{i=1}^{N-1} V_{i,j}^2 + \tfrac{1}{2} V_{N,j}^2 \right) \le h^2 N^2 S_x(V) = S_x(V).$$

The same pointwise bound with $i = N$ yields $\|V_B\|_{\Gamma,h}^2 = h\sum_j V_{N,j}^2 \le hNS_x(V) = S_x(V)$. Finally (5.3) bounds $S_x(V)$ by the energy. □

Proof of Proposition 5.3. Let w_c be the boundary weight in (5.1), so that $\gamma_c = hw_c$. The equation $A_h^c \boldsymbol{V} = \boldsymbol{d}$, where $\boldsymbol{d}$ has entries F and S, gives

$$\|V\|_{E,h,c}^2 = h^2 \sum_{\mathcal{I}_h} V_{i,j} F_{i,j} + h^2 w_c \sum_{j=1}^{N-1} V_{N,j} S_j.$$

Cauchy–Schwarz and Lemma 5.2 imply

$$\begin{aligned} \|V\|_{E,h,c}^2 &\le \|V_I\|_{I,h} \|F\|_{I,h} + h w_c \|V_B\|_{\Gamma,h} \|S\|_{\Gamma,h} \\ &\le \|V\|_{E,h,c} (\|F\|_{I,h} + \gamma_c \|S\|_{\Gamma,h}). \end{aligned}$$

Here $\|V_I\|_{I,h} \le \|V\|_{2,h} \le \|V\|_{E,h,c}$. If the energy norm is zero the claim is immediate; otherwise division by it proves (5.6). □

Proof of Corollary 5.4. Apply (5.6) to the error equation with $F = -\tau_I$ and $S = -\tau_B^c$. Since $h^2(N-1)^2 \le 1$ and $h(N-1) \le 1$, the restricted norms satisfy $\|\tau_I\|_{I,h} \le \|\tau_I\|_{\infty,I}$ and $\|\tau_B^c\|_{\Gamma,h} \le \|\tau_B^c\|_{\infty,B}$. Insert (4.6). The boundary factors are 1 for FO and $h/2$ for G, giving the two energy bounds exactly as stated. Lemma 5.2 supplies the L^2 bounds. □

B Supplementary numerical data

Source version and reproduction scope

These appendices accompany the repository snapshot 20ae3f7 of 22 October 2026. Tables in this appendix are rounded directly from the committed CSV files; the files retain their stored decimal precision. N/FO, N/G, R/FO and R/G denote Neumann or Robin data with the backward-derivative or ghost-eliminated closure. The mixed-boundary refinement set is $N = 16, 32, 64, 128, 256, 512$, with $h = N^{-1}$.

The archived run used Windows 11, Python 3.13.5, NumPy 2.1.3, SciPy 1.15.3, Matplotlib 3.10.0 and scikit-fem 12.0.2, as recorded in `data/run_metadata.json`. This metadata, rather than the different Python version stated in the README, identifies the saved run. The sparse direct baseline is SuperLU through `spsolve(..., use_umfpack=False)` and has no user-set stopping tolerance. Timings belong to that archived machine and run.

A distinct AI-assisted verification was run on 22 October 2026 on Windows 11 with Python 3.12.14. It passed sixteen unit tests and checked twelve small-grid asymmetric manufactured-solution configurations. The maximum defect discrepancy was 4.775×10^{-12}. The archived CSV files and figures were not overwritten. This run did not repeat the twenty-four large-grid refinement cases, energy tables or timings. Its plan, script and outputs are retained in `appendix_evidence.md` and `audit/`; it is separate from the archived run and the students' handwritten checks.

File	Rows	Evidence retained
data/refinement_results.csv	24	Nodal errors, residuals, nonzeros and measured costs
data/energy_convergence.csv	24	Energy errors and edge-sum identity checks
data/fitted_orders.csv	4	Fits and error-magnitude Richardson estimates
data/weighted_matrix_diagnostics.csv	4	Weighted structure on $N = 32$
data/condition_estimates.csv	16	Conditioning diagnostics on $N = 16, 32, 64, 128$
data/algebraic_error_separation.csv	12	ILU–GMRES tolerance sweep on $N = 64$
data/weak_gmres_comparison.csv	12	Jacobi–GMRES tolerance sweep on $N = 64$
data/cg_comparison.csv	4	Weighted CG on $N = 64$
data/fem_oracle_comparison.csv	24	Independent P1 FEM comparison on six meshes
data/pure_neumann_depth.csv	10	Separate all-flux problem on five meshes
data/cost_accuracy_candidates.csv	32	Eight candidate grids in each configuration
data/matched_accuracy.csv	20	Five targets in each configuration
data/error_locations.csv	20	Interior maximisers on $N = 32, 64, 128, 256, 512$

Table 14: Clickable source files at the fixed repository snapshot.

Interpretation of conditioning estimates

The exact-arithmetic inequality is

$$\|U - U_h\|_\infty \le \|A^{-1}\|_\infty \|b - AU\|_\infty, \qquad AU_h = b.$$

The CSV field `forward_error_bound_inf` substitutes a computed one-norm estimate for $\|A^{-1}\|_\infty$. SciPy 1.15.3 documents `onenormest` as a lower norm estimate. Consequently that substitution is a conditioning-based indicator, not automatically a certified upper bound for an arbitrary matrix.

Proposition B.1 (Inverse norm under inverse positivity)**.** *If A is invertible and A^{-1} is entrywise nonnegative, then*

$$\|A^{-1}\|_\infty = \|A^{-1}\mathbf{1}\|_\infty.$$

Proof. The infinity norm is the largest absolute row sum. Since every entry of A^{-1} is nonnegative, its row sums equal the components of $A^{-1}\mathbf{1}$. Taking their maximum proves the identity. □

The report's inverse-positivity result makes this identity applicable to the mixed systems. In the additional replay, independent solves of $Av = \mathbf{1}$ on $N = 12,128$ in all four configurations agreed with the norm estimator within 5.855×10^{-14} relatively. This supports its numerical accuracy on those checked systems; it is not an interval-arithmetic certificate for all refinements. The archived conditioning file covers $N = 16, 32, 64, 128$, not $256, 512$. The separately obtained solver comparisons provide additional evidence on their stated representative mesh.

C Pre-computation predictions and reconciliation

This appendix follows the instructor's *Questions Before Calculation* worksheet: prediction, output and reconciliation. No dated original prediction sheet was found in the inspected repository history. The first table therefore records analytical expectations reconciled *retrospectively* with the archived run; it does not claim that these entries were written before that run. A genuinely recorded before/after sequence for the additional replay follows separately.

Archived study: analytical expectations and retrospective checks

Item	Prediction / analytical expectation	Output	Reconciliation decision
1	For the exact benchmark, $1 \le u^* \le e^2$, $u^*_x = u^*_y > 0$, $f = -2u^* < 0$, and $s_\beta = (1+\beta)e^{1+y} > 0$ on the right edge.	The source and boundary functions implement these expressions; the normal/ghost test and manufactured checks pass.	Retain the outward normal $+x$ on the right edge. The model signs concern the exact solution; a small algebraic residual does not verify the PDE signs by itself.
2	No grid-matrix symmetry follows from the appearance of e^{x+y}. The raw A is nonsymmetric; $K = WA$ is SPD under the proved assumptions.	At $N = 32$, all raw symmetry defects are positive, all weighted defects are zero, and estimated smallest eigenvalues are 11.61–14.87.	Use SuperLU or GMRES on A and CG only on $KU = Wb$. Numerical eigenvalues support the implementation; the proof supplies SPD.
3	The smooth-problem theory gives $E_\infty = O(h)$ for FO and $O(h^2)$ for G; both raw boundary defects are $O(h)$.	Maximum-error fits on $128, 256, 512$ are $1.00491, 1.99993$ (Neumann) and $1.00253, 1.99994$ (Robin).	The observed rates agree with the stated bounds. The ghost stability estimate contributes the extra factor h. No matching G lower bound or general corner-singular rate is inferred.
4	$n = N(N-1)$; $m_{\mathrm{FO}} = 5N^2 - 11N + 6$ and $m_{\mathrm{G}} = 5N^2 - 9N + 2$.	All twenty-four saved cases and their fresh replays satisfy these counts. At $N = 512$, $n = 261632$ and $m = 1305094$ or 1306114.	The sparse layout agrees with the derivation. The extra 1020 G entries are about 0.078% of the FO count.

Item	Prediction / analytical expectation	Output	Reconciliation decision
5	Affine functions are stencil-exact for both closures. For $1 + x^2 + y^2$, G is exact and the FO boundary defect is $-h$.	The independent manufactured checks pass. At $N = 512$, G/FEM nodal differences are 3.090×10^{-6} (Neumann) and 1.229×10^{-6} (Robin).	Distinguish exact polynomial relations from asymptotic agreement of two different discretisations; FEM is not an exact discrete FDM solution.
6	In this manufactured family, changing β while regenerating s_β leaves the exact u^* unchanged. The limit $\beta = 0$ is Neumann.	Both boundary variants converge to the same exact function.	Interpret changes as closure/parameter effects on discretisation, not as a physical monotonic response at fixed Robin loading.
7	A solver cross-check is accepted only when $\rho_{\text{alg}} = \|U_{\text{it}} - U_{\text{direct}}\|_\infty / E_\infty < 0.01$. A library success flag alone is insufficient.	With Jacobi–GMRES and $\text{rtol} = 10^{-8}$, the two G ratios are 0.07402 and 0.06139. At 10^{-10} they are 5.471×10^{-4} and 7.475×10^{-4}.	Reject the loose-tolerance G runs as accuracy cross-checks despite their zero exit status. The tighter runs satisfy the stated criterion; this is a solver-accuracy issue, not a failure of the PDE model.
8	No theorem here predicts the exact location of a maximum error, a universal speed-up, or equality of all-flux compatibility defects.	Saved interior maximisers lie at $i = N - 1$ and their heights move with refinement. Finite cost targets can be unattained; pure-Neumann FO and G compatibility defects differ.	Treat these as observations with their stated grids and settings. They are not retrospectively relabelled as prior predictions.

Table 15: Expectation–output–decision record for the archived study; the chronology of these entries is explicitly retrospective.

The exact exponential function is invariant under exchanging x and y, but the boundary decomposition is not. Thus that one benchmark can conceal a coordinate-transposition error. The additional asymmetric test below was selected to address this risk directly.

Additional verification: a new recorded replay

The evidence below belongs to a separate verification run on 22 October 2026. Its plan was saved at `09:48:08 UTC`; execution ran from `09:49:12 UTC` to `09:49:14 UTC`. The archived results were already known when this new plan was written, so this is a prospective record for the new run, not a blind prediction of the original experiment.

The source was checked against repository commit `20ae3f7`. The new run used Windows 11, Python 3.12.14, NumPy 2.5.4, SciPy 1.18.1 and scikit-fem 11.0.0. These versions differ from the archived numerical run; no archived timing or byte-for-byte output reproduction is claimed. Before the new execution, the fixed asymmetric manufactured solution was

$$v(x, y) = 2 + x + 2y + 3x^2 + xy + 2y^2, \qquad -\Delta v = -10,$$

$$v(1, y) = 6 + 3y + 2y^2, \qquad \partial_n v(1, y) = 7 + y, \qquad s^v_\beta = 7 + y + \beta(6 + 3y + 2y^2).$$

For $N = 6, 12, 24$, two boundary types and two closures, the plan specified twelve configurations. Centred differences predict zero interior and ghost-row defects up to round-off. The backward derivative underestimates $v_x(1, y)$ by $3h$, giving the FO boundary defect $-3h$. The pre-run acceptance limits were 10^{-9} for the maximum defect discrepancy and ghost nodal error; all sixteen unit-test methods also had to pass. A convergence slope is not fitted to round-off-sized errors.

Check	Observed in the new run	Decision
Manufactured configurations	12/12 completed	Pass
Maximum defect discrepancy	4.77485×10^{-12}	$< 10^{-9}$
Maximum G nodal error	3.10862×10^{-14}	$< 10^{-9}$
Unit tests	16/16 passed	Pass

Table 16: Predictions and outputs for the distinct 22 October verification run.

The pre-run plan's SHA-256 was unchanged before and after execution; the full digest is recorded in the supporting evidence file. The command, source checks, script, result JSON and complete test output are described in `appendix_evidence.md` and retained under `audit/`. This new verification did not rerun the archived twenty-four large-grid refinement cases, energy tables or timings. It is AI-assisted technical evidence, separate from the team's handwritten five-gate review.

D AI Transparency Log

Part A — Team AI Transparency Log

A1. AI tools used

Tool	Version / Date	Primary use
ChatGPT / Codex	Exact historical model/version not recorded; this appendix audit: 22 October 2026.	English and LaTeX drafting, mathematical explanations and revisions, reference checking, proposed code changes and automated verification.

Routine language and formatting assistance is summarised once here. Any additional AI tools actually used by team members must be added by their users. Numerical libraries are trusted dependencies, rather than AI tools, and are identified in A4.

A2. Significant AI-assisted claim audits

Item	Prompt / claim / assumptions	Risk and possible falsifier
1	Derive FO/G boundary rows and justify their global orders for the smooth square-domain problem, fixed $\beta \geq 0$ and exact Dirichlet data.	A centred derivative alone does not prove a second-order solution. A wrong sign, missing $2s_\beta/h$ load or invalid stability weight would falsify the derivation.
2	Decide whether the raw matrix or a weighted system is eligible for CG. The claim concerns $K = WA$ with the specified weights and $N \geq 2$.	Raw nonsymmetry can be hidden by the self-adjoint PDE. A mismatched weighted pair or nonpositive energy would falsify SPD.
3	Verify assembly, source-to-table consistency and solver accuracy for the fixed repository version.	A symmetric benchmark can conceal transposed coordinates; two solves can share a wrong matrix; a zero exit status or a norm estimate can be mistaken for an accuracy certificate.

A3. Independent verification and decision

Item	Independent check and evidence location	Final decision / modification / limitation
1	Exact ghost elimination and truncation/stability proofs in report Sections 3–5 and Appendix A; analytic defects in `code/tests/test_independent_assembly.py:54--68`; all six-mesh errors in Table 7.	Retain the conditional FO $O(h)$ and G $O(h^2)$ upper bounds. The FO matching lower bound is benchmark-specific; no matching G lower bound is claimed.
2	Coefficient pairing and the edge-sum proof; weighted tests in `code/tests/test_mixed_poisson.py:75--90`; `data/weighted_matrix_diagnostics.csv` and `data/cg_comparison.csv`.	CG is applied to $KU = Wb$, with positive Jacobi preconditioning. The proof establishes SPD; sampled eigenvalues are implementation checks.
3	Independent uneliminated dense assembly, P1 FEM, the asymmetric polynomial, saved-data checks, the sixteen-test replay and the tolerance failures recorded in Appendix C.	Accept the recorded technical evidence at this source version. Reject loose G/Jacobi–GMRES as a 1% accuracy check; label estimated conditioning quantities honestly. The replay does not replace independent team review or signatures.

The alternate assembly and FEM provide different mathematical routes. FEM shares the continuum data and sparse linear-algebra environment; it is not an exact FDM oracle or an independently authored human audit. The current replay was executed with AI assistance. Team members must retain their own reproduction and review records.

A4. Intellectual ownership declaration

By signing, each member affirms that they can explain and modify the team-owned discretisation and solver code, justify the mathematical claims, and state the assumptions and interface tests at trusted libraries without AI assistance during the viva. A small live change or diagnostic may be required. An unsigned row is not a completed declaration.

Name	Signature	Team-owned artefact / contribution	Trusted dependency and interface
Kainan Zhao		PM, mathematical exposition, report integration and references.	Sparse-system assumptions, matrix weighting and interpretation of solver diagnostics.
Suning Peng		`code/models.py`, experiment execution and automated testing.	SciPy sparse storage, SuperLU and Krylov interfaces; dimensions, status and original-system residuals.
Rengerile Na		Validation of numerical evidence, figure preparation and interpretation.	NumPy norm calculations, Matplotlib plots and the scikit-fem comparison; common-node mapping and source data.

Five-gate solver audit

This section follows *Auditing an AI-Generated Solver*. The inspected numerical implementation is `code/models.py:92--168` and the associated experiment/test code at commit `20ae3f7`. The additional technical audit was executed on 22 October 2026 by ChatGPT/Codex, after the archived results existed. It does not certify that the historical code was audited before use. Student verifier names and signatures remain separate from AI execution.

Gate	Finding	Evidence (file:line, output)	Action taken	Verified by
1	The PDE, data and two closures match the stated mixed problem.	`code/models.py: 25--35,92--147`; explicit rows below; analytic and asymmetric manufactured defects.	Check RHS signs, outward normal, $\beta = 0, 2$, and the G load $2s_\beta/h$ against the report. Technical checks satisfied in the replay.	Suning Peng.
2	Unknown ordering, corner ownership, sparsity and nonstandard rows agree.	`code/models.py: 38--46,104--156,172--181`; all twenty-four degree/nonzero counts; independent uneliminated systems.	Spot-check both ends and an interior index; verify row coefficients and eliminated corner loads. Technical checks satisfied.	Suning Peng.
3	Nonlinear Jacobian testing is not applicable. The residual is $F(U) = AU - b$, so $DF(U) = A$.	No Newton loop is present. Direct solve: `code/models.py:160--168`. Krylov status checks: `code/experiments.py: 435--587,652--697`.	Mark the nonlinear gate N/A. Retain explicit tolerances, iteration caps, exit status and recomputed residuals for linear cross-checks.	Kainan Zhao.
4	Exact data, a different assembly and a different discretisation support the results.	`code/tests/test_independent_assembly.py:13--68`; `code/fem_oracle.py:35--93`; sixteen tests pass in the new verification; twelve asymmetric small-grid configurations pass. The archived twenty-four refinement cases were not rerun in this verification.	Keep the exact reference separate from assembly; compare dense uneliminated and reduced systems; compare P1 FEM at shared nodes. FEM differences are discretisation differences, not zero-tolerance equalities.	Rengerile Na.
5	Three failure hypotheses were posed and tested below. One tolerance failure was reproduced; an estimated-bound interpretation was restricted.	Asymmetric quadratic, Jacobi–GMRES sweep, inverse-positive norm identity and replay evidence.	Retain the failures and limitations. Team members must independently select, audit and explain these checks; no viva outcome is asserted.	Rengerile Na.

Table 17: Audit record using the instructor's five required columns.

Explicit boundary rows and index checks

Before Dirichlet elimination, the FO right-edge row is

$$(h^{-1} + \beta)U_{N,j} - h^{-1}U_{N-1,j} = s_\beta(y_j).$$

The G row, obtained by eliminating

$$U_{N+1,j} = U_{N-1,j} + 2h(s_\beta - \beta U_{N,j}),$$

is

$$\left(\frac{4}{h^2} + \frac{2\beta}{h}\right)U_{N,j} - \frac{2}{h^2}U_{N-1,j} - \frac{U_{N,j-1} + U_{N,j+1}}{h^2} = f_{N,j} + \frac{2s_\beta(y_j)}{h}.$$

At $N = 4$, $j = 1$, the G retained coefficients are $64 + 8\beta$, -32 and -16 at $(4,1), (3,1), (4,2)$. The eliminated $(4,0)$ corner adds $16e$ to the load:

$$b_{k(4,1)} = (6 + 8\beta)e^{5/4} + 16e.$$

The corresponding FO coefficients are $4 + \beta$ and -4, with load $(1 + \beta)e^{5/4}$ and no tangential boundary couplings. The zero-based map is

$$k(i,j) = (j-1)N + i - 1.$$

For $N = 4$, $(1,1) \mapsto 0$, $(2,2) \mapsto 5$ and $(4,3) \mapsto 11$. Dirichlet nodes are excluded, and reconstruction uses $U[j,i]$.

Gate 5: three ways convergence alone could be misleading

1. **A symmetric reference conceals coordinate transposition.** The exponential is unchanged by exchanging x and y. The asymmetric quadratic in Appendix C gives different directional coefficients and an exact FO defect $-3h$. Its twelve checks pass the predeclared thresholds.

2. **A successful solver flag conceals excessive solve error.** In the archived Jacobi–GMRES G results, the 10^{-8} run terminates with status zero but fails the one-percent criterion, whereas the 10^{-10} and 10^{-12} runs satisfy it. A separate 22 October audit checked 10^{-6} and 10^{-12}: both reported successful exit, but the former failed and the latter passed the one-percent algebraic-error budget. This later check demonstrates the same risk; it is not a rerun of the archived tolerance sweep.

3. **A norm estimate is mistaken for a rigorous forward bound.** The CSV field name `forward_error_bound_inf` does not alter the estimator's mathematical guarantee. Proposition B.1 provides a second evaluation route for the inverse-positive matrices. The checked values agree numerically, but the indicator remains distinguished from a verified upper-bound certificate.

Student reviewers and actual dates. The team reports that Suning Peng checked Gates 1–2, Rengerile Na checked Gates 4–5, and Kainan Zhao checked Gate 3, all on 22 October 2026. The reviewer labels in `audit/five_gate_plan.json` reflect an earlier allocation. The plan is retained unchanged for hash verification; the allocation stated above records the team's actual reviewers and changes no numerical result.

Reproduction methods and retained evidence. The hand-checkable boundary-row derivation is in `audit/boundary_rows_review.md`; the independent-assembly and Gate 5 review worksheets are in `audit/independent_checks_review.md` and

`audit/gate5_report_alignment.md`. These are typeset review materials, not handwritten scans. From the exact source snapshot at commit `20ae3f7`, the new technical checks can be rerun with `python audit/run_five_gate_checks.py` and `python audit/inverse_norm_followup.py`. Their plans, scripts and raw results are retained under `audit/`; the sixteen-test terminal output is in `audit/runs/five_gate/unit_tests.txt`. The run environment and numerical outcomes are summarised in `five_gate_evidence.md`.

Decision and limits. The new checks found agreement in twelve $N = 4$ right-edge rows, sixteen independent small-system solves, and a decreasing P1 FEM reference error; all sixteen unit-test methods passed. A loose GMRES setting failed the declared algebraic-error budget despite successful exit, while a tight setting passed. The inverse-norm routes agreed on eight small matrices, but their agreement does not certify an estimator as a universal upper bound. The archived twenty-four large-grid cases and timings were not reproduced by these new checks.

Source and interface links

The instructor-supplied *Auditing an AI-Generated Solver*, *Questions Before Calculation*, and `ICS_bvp.tex` determine the audit, reconciliation and declaration formats. No public URLs were supplied for those handouts. The following links identify the implementation and the exact versioned library interfaces used in this appendix:

- Project repository at the audited commit.
- SciPy 1.15.3: `onenormest`.
- SciPy 1.15.3: `spsolve`; `gmres`; `cg`.
- scikit-fem 12.0.2 source release.

E Individual Contribution Statements

Part B — Kainan Zhao

Individual Contribution Statement — Project 2 (PDE BVP)

Name: Kainan Zhao
Role this project: Project Manager, Mathematical Theory, and Report Writing (roles 1 and 2, and the writing component of role 4).

The specific things I did this project:

1. Coordinated the division of responsibilities between analysis, implementation, experiments, validation and visualisation. I organised the report around the selected mixed-boundary problem and linked the mathematical discussion with the numerical evidence required from the other members.

2. Prepared and revised the mathematical exposition in the Overleaf manuscript, particularly Sections 2–5. This covered the PDE and data, corner ownership, FO/G rows, consistency, stability and convergence, with lemma/proposition-based proofs. The AI-assisted development and revision of these arguments is disclosed in Part A, Items 1 and 2.

3. Integrated the English LaTeX report, introduction, conclusions and linked references. I organised reviewer-driven corrections and worked to distinguish theoretical bounds, observed rates and limits of the numerical evidence. The current report artefact is `report/MTH321_project2.pdf`; its complete editable source is maintained in Overleaf.

What I would do differently:

I would establish shared notation and a source-to-table checklist earlier, so that changes to the code, data and figures could be reflected consistently in the manuscript.

Anything I should have asked for help with earlier:

I should have clarified the experimental source version and the distinction between numerical checks, a genuine pre-computation record and an independently signed team audit earlier.

Part B — Suning Peng

Individual Contribution Statement — Project 2 (PDE BVP)

Name: Suning Peng
Role this project: Algorithm Implementation, Numerical Experiments and Automated Testing (role 3 and the experiment/test-execution component of role 5).

The specific things I did this project:

1. Worked on the finite difference implementation in `code/models.py`, including `assemble_system`, `solve_system` and `reconstruct`. My responsibilities covered the index map, vectorised sparse assembly, Dirichlet elimination and the two right-edge closures for Neumann and Robin data.

2. Worked on experiment execution through `code/run_all.py` and `code/experiments.py`, including `run_refinement` and the solver-comparison routines. I organised errors, residuals, matrix diagnostics, timings and exported CSV outputs for the validation and visualisation work.

3. Handled automated testing in `code/tests/`, including `test_mixed_poisson.py` and `test_independent_assembly.py`. The implementation checks cover dimensions, boundary rows, corners, manufactured defects, invalid inputs and convergence. Part A records the relevant AI-assisted revisions and distinguishes the additional appendix replay from the team's original experiment execution.

What I would do differently:

I would settle the output format and benchmarking settings before the full experiments, and keep assembly, solving and diagnostic work clearly separated in the recorded cost measure.

Anything I should have asked for help with earlier:

I should have discussed boundary-row scaling and the algebraic-error acceptance criterion with the theory and validation owners before finalising the solver comparisons.

Part B — Rengerile Na

Individual Contribution Statement — Project 2 (PDE BVP)

Name: Rengerile Na
Role this project: Validation and Numerical Visualisation (the validation component of role 5 and visualisation component of role 4).

The specific things I did this project:

1. Worked on validating numerical evidence against the manufactured solution. I focused on the error definitions in `code/models.py:compute_diagnostics` and the order calculations in `code/experiments.py:observed_orders` and `fit_loglog_order`, including the norms and fitting ranges used to interpret the two closures.

2. Reviewed the distinction between discretisation and algebraic error through the refinement and solver-comparison CSV files in `data/`. My validation responsibilities included boundary and corner behaviour, weighted symmetry and interpretation of residuals and tolerance failures. Part A, Item 3, identifies additional AI-run checks separately from personal contribution records.

3. Worked on the numerical figures and their interpretation through `code/make_figures.py`, including solution/error plots, matrix sparsity, refinement and cost-versus-accuracy comparisons. I focused on readable labels, reference slopes, legends and captions, and coordinated the explanation of the figures with Kainan.

What I would do differently:

I would agree the norms, fitting ranges and plotting conventions before the full experiments, and maintain an index linking each figure to its CSV source, code version and solver settings.

Anything I should have asked for help with earlier:

I should have requested a clearer account of the settings behind each exported dataset, especially when interpreting timing results and the limitations of conditioning estimates.